INCH-POUND

MIL-S-43926J
30 August 1991
SUPERSEDING
MIL-S-43926H
25 March 1988

MILITARY SPECIFICATION

SUIT, CHEMICAL PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable suit for wear as an overgarment to protect against chemical warfare agents for periods of 1 to 22 days. The suit consists of separate coat and trousers and is made with a cotton/nylon shell and a sewn in laminated foam lining impregnated with activated charcoal. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The suit shall be of one type in the following classes and sizes (see 6.2).

Class 1 - Woodland Camouflage
Class 2 - Deleted (see 6.7)
Class 3 - Desert Camouflage (3 color)

Schedule of sizes

XXX-Small	Medium	Large
XX-Small		X-Large
X-Small		XX-Large
Small		

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-50198	-	Thread, Gimp, Cotton Buttonhole
A-A-50199	-	Thread, Polyester Core, Cotton- or Polyester-Covered
V-F-106	-	Fasteners, Slide, Interlocking
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing
MIL-B-371	-	Braid, Textile, Tubular
MIL-B-543	-	Buckles, Tongueless and Web Strap
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-T-3530	-	Thread and Twine, Mildew Resistant or Water Repellent Treated
MIL-F-10884	-	Fasteners, Snap
MIL-C-12189	-	Cloth, Coated: Butyl Coated, Toxicological Agents Protective
MIL-G-16491	-	Grommet, Metallic, General Specification for
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43303	-	Cord, Elastic, Cotton
MIL-T-43566	-	Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
MIL-C-43858	-	Cloth, Laminated, Nylon Tricot Knit, Polyurethane Foam Laminate, Chemical Protective and Flame Resistant

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

MIL-C-44031 - Cloth, Camouflage Pattern: Woodland, Cotton and Nylon
MIL-C-44034 - Cloth, Twill, Camouflage Pattern, Cotton and Nylon For Desert Uniform
MIL-P-44406 - Pouches, Plastic Film

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials
FED-STD-595 - Colors Used in Government Procurement
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Test Method for Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Outer shell. The camouflage printed and quarpel treated cotton and nylon twill cloths used for the outer shell for the classes 1 and 3 suits shall conform to the requirements specified in 3.3.1.1 and 3.3.1.2.

3.3.1.1 Class 1 suit. The cloth shall be woodland camouflage printed and shall conform to class 2 of MIL-C-44031, except that the colorfastness to laundering (after three cycles) shall rate at least "fair".

3.3.1.2 Class 3 suit. The cloth shall be desert camouflage printed (3 color) and shall conform to class 3A of MIL-C-44034.

3.3.2 Lining. The laminated cloth used for the lining shall conform to type III of MIL-C-43858 (see 6.5).

3.3.3 Butyl cloth. The butyl cloth used for protective pieces beneath all coat and trouser pockets and as reinforcement for elbows and knees shall conform to MIL-C-12189. The heavily coated side of the butyl cloth, when used, shall face to the outside of the coat or trousers. The color shall approximate Olive Drab 177. The colorfastness and dry crocking requirements shall not apply. In addition, the adherability test for lb/inch width, initial, after water immersion, and after heat aging requirements of MIL-C-12189 shall not apply. Initial toxicological agents resistance shall meet the requirements specified in MIL-C-12189 when tested in accordance with 4.4.1.1.

3.3.4 Elastic cord. The cotton elastic cord for the drawcord at bottom of coat shall be 3/16 inch in diameter, conforming to class 2 of MIL-C-43303. The color shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Tan 380 for class 3 suit, and the colorfastness requirements shall not apply.

3.3.4.1 Elastic cord resin impregnating agent. The resin agent used for dipping or impregnating the ends of the drawcords shall be cellulose acetate or cellulose acetate butyrate to prevent ravelling. The dipped or impregnated ends shall be at least 1/2 inch in length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

3.3.5 Slide fasteners. The slide fasteners shall be continuous element nylon or polyester (spiral type), size MS and in the types and styles indicated. The slide fasteners shall conform to the requirements of V-F-106 before being sewn into the garment. The salt spray test shall not apply. After insertion into the garment, the pin must be able to be fully inserted through slider, seated into box and operable. The closed-end bottom stop shall be a staple type and shall be positioned and sewn, enclosing the sides and ends of the fastener chain. Solid type closed-end bottom stops shall abut the end of the slide fastener chain, and be securely affixed to the slide fastener tape without damage to the tape or chain. The slide fastener pulls shall have a hole with sufficient size to accommodate a 3/8 inch wide thong when a short tab pull is used on the coat front, the tab shall have a minimum length of 7/8 inch.

Coat - Type IV, style 8, long or short tab pull
Trouser fly and leg openings - Type I, style 3 or style 4 (cam only) short tab pull

3.3.5.1 Slide fastener tape. The color of the slide fastener tape shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Khaki Shade P, Cable Number 66516 for class 3 suit, and the colorfastness requirements shall not apply. The width of the slide fastener tape shall be 9/16 or 5/8 inch.

3.3.6 Braid.

3.3.6.1 Leg drawcord. The solid tubular cotton braid for drawcords, at leg bottoms shall conform to type IV, class 2, 4/32 inch diameter of MIL-B-371. The color shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Tan 380 for class 3 suit, and the colorfastness requirements shall not apply.

3.3.6.1.1 Drawcord braid resin impregnating agent. The resin agent used for dipping or impregnating the ends of the drawcords shall be cellulose acetate or cellulose acetate butyrate to prevent ravelling. The dipped or impregnated end shall be at least 1/2 inch in length.

3.3.6.2 Suspender loops and slide fastener pull thongs. Either braid conforming to type VII of MIL-B-371, or outer shell material (see 3.3.1) may be used for these components. The color shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for the class 1 suit, and Tan 380 for class 3 suit, and the colorfastness requirements shall not apply. Cutting and fabrication details when outer shell material is used shall be as specified in table I.

3.3.7 Hook and loop fastener tape. The hook and loop fastener tape for securing coat chest pockets, and coat center front overlap shall be 5/8 inch wide hook and loop tape, and for sleeve cuff adjustments, coat pocket flaps, trouser hip pocket flaps and hip pockets shall be 1 inch wide hook and loop

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

tape conforming to type II, class 1 of MIL-F-21840. The color shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Tan 380 for class 3 suit, and the colorfastness requirements shall not apply.

3.3.8 Tape.

3.3.8.1 Adjustment strap tapes. The adjustment strap tape shall be type I, class 3, 5/8 inch width, Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Tan 380 for class 3 suit, conforming to MIL-T-43566, except that the colorfastness requirement shall not apply.

3.3.8.2 Chest pocket pull tapes. The chest pocket pull tapes, when not cut from ends, shall conform to type I, class 3, 3/8 inch width, Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Tan 380 for class 3 suit, conforming to MIL-T-43566, except that the colorfastness and curvature requirements shall not apply.

3.3.9 Snap fasteners. Snap fastener assemblies conforming to MIL-F-10884 shall be used to secure front closure of trousers, to attach the coat back to the trouser back, and to secure trouser cargo pocket flaps. The fasteners to secure trouser closure and trouser cargo pocket flaps shall conform to style 2 or 2A, finish 2. Style 2 shall consist of a 24 line button, size 1 or 2 socket for the female half; and eyelet size 1 or 2, with stud for the male half. Style 2A shall consist of a button, size 1 or 2 and socket for the female half; and eyelet, size 1 or 2, with stud for the male half. The following exception to MIL-F-10884 shall apply: A double stud or as an alternate, style 2 or 2A stud-eyelet combination shall be used on the right side of trousers at waist opening. The snap fasteners for attaching the back of the coat to the trousers shall conform to style 2, finish 2, 24 line size button. The barrel length of button and eyelet shall be selected by the contractor so as to insure a secure clinch of components to material.

3.3.9.1 Grommet. The grommets (when applicable) for the coat bottom and trouser leg drawcords shall conform to type II, class 3, size 0 of MIL-G-16491.

3.3.10 Buckles. The buckles for hip adjustment straps shall be tongue-less, 2 bar, conforming to type II, class 3, style 3, 5/8 inch of MIL-B-543.

3.3.11 Gimp. The cotton gimp for reinforcing buttonholes shall conform to type I or II, ticket number 8 of A-A-50198 and the color shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Khaki P-1, Cable No. 66019 for class 3 suit, and the colorfastness requirements shall not apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

3.3.12 Thread, cotton- or polyester-covered, polyester core. The cotton-covered or polyester-covered, polyester core threads for seaming and stitching the coat and trousers shall be ticket numbers 50 and 70, 2 ply conforming to A-A-50199.

3.3.12.1 Color. The threads shall be Camouflage Green 483 approximating Color Chip 34094 of FED-STD-595, for class 1 suit, and Khaki P-1, Cable No. 66019 for class 3 suit, and the colorfastness requirements shall not apply. The thread used to sew the laminate cloth for the lining may be white where not exposed to outside.

3.3.12.2 Water repellent treatment. The cotton-covered or polyester-covered polyester core threads shall be water repellent treated conforming to type II, class 3 of MIL-T-3530. Lubrication of the quarpel treated thread by any means prior to or during the sewing is prohibited.

3.3.13 Labels. All clothing labels shall be in accordance with DDD-L-20. Each of the coats and trousers shall have an identification, a size, and an instruction label. These three labels may be combined. The colorfastness requirements shall not apply.

3.3.13.1 Identification label. The identification label shall conform to type VI, class 1 of DDD-L-20, except that the surveillance marking requirements in MIL-C-43858 for the laminated cloth shall appear under the last line of inscription on the label. The surveillance marking may be hand printed or stamped. Size of characters for the lot number shall be a minimum of 10 points affixed with Black (No. 37038) ink conforming to type I of A-A-208.

3.3.13.2 Size label. The size label shall conform to type VI, class 2 of DDD-L-20.

3.3.13.3 Instruction label. The instruction label shall conform to type VI, class 3 of DDD-L-20 and shall include the following instructive information for each respective size. Instructive information shall be printed in letters not less than a minimum 10 point (approximately 1/8 inch).

1. Wear as outergarment layer of clothing for protection against chemical agents. Should not be worn with underwear only, except in very hot environment.

Number 2 shall be as follows for each separate size:

For size XXX-Small:

2. The multisize system permits the XXX-Small to be worn by the size XX-Small person when only the underwear is worn.

For size XX-Small:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

2. The multisize system permits the XX-Small to be worn by the size X-Small person when only the underwear is worn.

For size X-Small:

2. The multisize system permits the X-Small to be worn by the size Small person when only the underwear is worn.

For size Small:

2. The multisize system permits the Small to be worn over the X-Small cold-wet uniform or the Small hot weather uniform or by the size Medium person when only the underwear is worn.

For size Medium:

2. The multisize system permits the Medium to be worn over the X-Small cold-dry uniform, the Small cold-wet uniform or the Medium hot weather uniform or by the Large person when only the underwear is worn.

For size Large:

2. The multisize system permits the Large to be worn over the Small cold-dry uniform, the Medium cold-wet uniform or the Large hot weather uniform or by the X-Large person when only the underwear is worn.

For size X-Large:

2. The multisize system permits the X-Large to be worn over the Medium cold-dry uniform, the Large cold-wet uniform or the X-large hot weather uniform.

For size XX-Large:

2. The multisize system permits the XX-Large to be worn over the Large cold-dry uniform or the X-Large cold-wet uniform.

For Trousers

3. CAUTION: Before entering the water, when wearing the Chemical Protective Suit, the bottoms of the trouser legs should be untied and slide fasteners opened. This action will minimize accumulation of water between the suit and the body.

3a. CAUTION: Protection provided by the overgarment is completely degraded in the areas where the overgarment is wet through with fuel, oil or hydraulic fluid.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

4. Use suspenders, if required, for improved suspension.

5. In most circumstances conditions will permit you to remove your footwear prior to donning or doffing these trousers. However, in some conditions of chemical warfare it will be necessary to don and doff your trousers without removal of this footwear. In this event, care must be taken.

6. Inspect your leg closure system to insure it is free from dirt and ice, and is firmly compressed to provide adequate closure.

7. This suit is not launderable. It may only be laundered for training purposes.

For Coat

3. CAUTION: Before entering the water, when wearing the Chemical Protective Suit, the slide fastener should be opened and the bottom untied. This action will minimize accumulation of water between overgarment and the body.

3a. CAUTION: Protection provided by the overgarment is completely degraded in the areas where the overgarment is wet through with fuel, oil or hydraulic fluid.

4. This suit is not launderable. It may only be laundered for training purposes.

DO NOT REMOVE THIS LABEL

3.3.13.4 Combination label. Size, identification, and instruction labels may be combined conforming to type VI, class 14 of DDD-L-20, except that the lot number of the laminated cloth shall appear under the last line of inscriptions on the label and may be hand printed or stamped. The characters for the lot number shall be a minimum of 10 points and shall be printed with Black (No. 37038) ink conforming to type I of A-A-208.

3.3.13.5 Fit and size prediction legend. The fit and size prediction legend shall be as specified below. The markings shall be applied to the vapor barrier bag in accordance with 5.4.2.

SUIT, CHEMICAL PROTECTIVE

(1) Check garment for proper fit. Use the Size Prediction Chart for your approximate size suit. Trade items as required.

(2) Issue suspenders simultaneously with suit for use if required to aid in suspension of trousers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

(3) These garments are items of outer wear and can be worn for up to 22 consecutive days. When exposed to chemical agent they will offer protection for at least 24 hours. At the end of 22 days or after 24 hours use in a chemical environment the overgarments are to be replaced.

(4) When maximum protection is required, wear this suit with Protective Mask and Hood; Glove Set, Chemical Protective; and Footwear Covers, Chemical Protective (Overboots).

(5) Use this prediction chart to determine the suit size you will need.

Size Prediction Chart

Waist size	Over underwear, long or short	Over battledress uniform	Over coat and trousers cold weather field wear	Over parka and trousers extended cold weather clothing
23	XXX-S	XX-S	X-S	S
27	XX-S	X-S	S	M
31	X-S	S	M	L
35	S	M	L	X-L
39	M	L	X-L	XX-L
43	L	X-L	XX-L	XX-L

3.4 Design.

3.4.1 Coat. The coat shall be a hip length design with stand-up collar and full length sleeves with adjustable tab and fastener tapes. Each sleeve shall have an outside one-sided bellows type pocket positioned below an inner atropine carrying pocket with one flap covering both pockets. The coat shall have two side-opening breast pockets, a slide fastener front closure with intermittent fastener tapes and an elastic drawcord in the hem. Each sleeve will have an impermeable elbow patch (see figure 1).

3.4.2 Trousers. The trousers shall have a slide fastener front opening with protective flap, two bellow type pockets with flaps located at the side, two hip pockets with flaps, hip adjustment buckles and straps, suspender loops and belt loops. Each leg shall have a slide fastener at the outside to close the gusset opening and a drawcord in the hem. Each leg will have an impermeable knee patch (see figure 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

3.5 Patterns. Standard patterns, which provide 1/2 +1/8, -0 inch seam allowance for double-lapped and double-stitched seams of coat and trousers, and 3/8 +1/8, -0 inch seam allowance for single needle seams, except where otherwise specified, will be furnished by the Government. The standard patterns shall not be altered in any way and are to be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns.

3.5.1 List of pattern parts. The components of the suit shall be cut from the materials as specified in accordance with the pattern parts indicated.

Materials	Nomenclature	Cut parts
	COAT	
Outer shell (cotton/nylon cloth)	Front	2
	Back	1
	Collar 1/	1
	Top sleeve	2
	Under sleeve	2
	Chest pocket	2
	Sleeve pocket	2
	Atropine pocket flap	2
	Atropine pocket facing shell	2
	Sleeve tab	2
Lining (laminate cloth)	Front	2
	Back	1
	Collar 1/	1
	Top sleeve	2
	Under sleeve	2
Butyl cloth	Top sleeve elbow patch	2
	Under sleeve elbow patch	2
	Chest pocket protective piece	2
	Atropine pocket	2
	Sleeve pocket protective piece	2
	TROUSERS	
Outer shell (cotton/nylon cloth)	Front	2
	Back	2
	Fly protective flap	1
	Hip pocket	2
	Hip pocket flap	2
	Leg slide fastener covers	4
	Cargo pocket facing	2
	Cargo pocket	2
	Cargo pocket flap	2

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:01Z
Check the source to verify that this is the current version before use.

Materials	Nomenclature	Cut parts
Lining (laminate cloth)	Front	2
	Back	2
	Fly protective flap	1
Butyl cloth	Knee patch	2
	Cargo pocket protective piece	2
	Hip pocket protective piece	2

1/ Collar shell and collar lining utilize same pattern part.

3.6 Construction.

3.6.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. Seam allowances shall be maintained so that no raw edges, runoffs, pleats, or open seams occur. When two or more methods of seams or stitch types are given for the same operation, any one of them may be used. When stitch type 401 is used, the looper (underthread) shall be on the inside of the coat and trousers. The gage for double-lapped and double-stitched seams shall be 1/4 to 5/16 inch.

3.6.1.1 Stitch tension. Prior to daily production of the chemical protective suits, the stitch tension of each sewing machine used in the fabrication of foam lining laminate cloth shall be pretested in accordance with the requirements of 3.6.1. Stitch tension on all lining (laminate cloth) seam operations shall be sufficient to compensate for the foam compression and support a 2.0 ounce weighted thickness gage when tested as specified in 4.5.4.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table I.

3.6.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch and not more than 1/2 inch. Thread breaks, all stitch types, shall be secured by stitching back of the break not less than 1/2 inch and not more than 3/4 inch. Skipped stitches or thread breaks, 401 stitch type, may be repaired using 301 stitch type. Ends of a continuous line of stitching shall overlap not less than 1/2 inch and not more than 3/4 inch.

3.6.4 Buttonholes and eyelets. The buttonholes shall be either straight cut type or the eyelet end taper bar type worked over gimp with the purling on the outside and the end of the buttonhole stitching and gimp securely

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

machine. The thread and stitches per inch shall be as specified in table I for the applicable operation.

c. Repairs on open seams in the joining seams of the lining shall be made by pinching the open seam together and stitching both plies 1/16 to 1/8 inch from end to end to form a dart tapered on each end. Shell material shall not be caught in the stitching and length of the repair shall not exceed 2 inches. Repairs are limited to two on the coat and two on the trouser.

d. When there are no more than two defects on the lining or the shell of any coat or trousers and when it is necessary to repair the shell or lining for an open seam in excess of 2 inches, the hem shall be opened and the repair made on the inside in conformance to the applicable operation in table I of the document.

3.6.6 Bartacking. Bartacking shall be 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.6.7 Setting of grommets. The holes shall be pre-punched to receive the grommets. The holes shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommet shall be securely clinched without cutting the adjacent material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

3.6.8 Setting of snap fasteners. When not using an automatic snap fastener setter, a hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.7 Manufacturing operations requirements. The coat and trousers shall be manufactured in accordance with operations listed in table I. Although the contractor is not required to follow the exact sequence of operations, the contractor is required to stitch the identification labels with surveillance marking already handprinted or stamped (see 3.3.13.1) to the lining (as specified in operations 12 and 41 of table I) immediately upon cutting to ensure correct identification and traceability of the cloth. Any basting or holding stitching may be used providing threads or needle holes are not visible on the finished garments. The coat and trousers shall be cut and assembled from the same lot of impregnated cloth.

3.7.1 Figures. Figures are furnished for information only. To the extent of any inconsistencies between the figures and the written specification, the written specification shall control.

3.8 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table I, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government furnished shaper pattern is forwarded, the component shall conform to that pattern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the coat and trousers in strict accordance with patterns furnished, which show size, directional lines and marks for proper assembly. Unless otherwise specified, the directional lines indicate the warp direction. b. Cut the outer shell (cotton/nylon cloth) parts from one lot of material except fly protective flap, belt loops, suspender loops, sleeve tabs, pocket facings, chest pocket pulls, atropine pocket, and slide fastener thongs which may be cut from ends. Parts cut from ends shall approximate the shade of the outer shell. NOTE: Pocket facings may be cut on rolls. c. Cut all parts of the lining (laminated cloth) from one lot of material. The knit side of laminated cloth is the face of the material (see 6.5.2). d. Directional lines on the patterns for front and back of trousers may vary from the warp direction by not more than 2-1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the directional line to the selvage edge of the fabric and the difference between the two measurements shall not exceed 2-1/2 inches. e. Cut the drawcord for leg bottom in 30 $\pm$ 1 inch lengths for all sizes.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) f. Cut the elastic braid for drawcord at bottom of coat in the following lengths (± 1 inch tolerance). Size — Inches (relaxed) XXX-Small — 38 XX-Small — 38 X-Small — 41 Small — 41 Medium — 44 Large — 44 X-Large — 47 XX-Large — 47 g. Cut the belt loops, suspender loops and slide fastener thongs to conform to the measurements specified in operations 4, 5, and 50. h. Cut the hook and loop portions of nylon fastener tape in the following lengths: Coat: Chest pocket (2 each of hook and loop) 6-1/2 ± 1/8 inches Sleeve pocket (4 each of loop) 1-1/2 ± 1/8 inches Atropine pocket flap (4 each of hook) 1-1/2 ± 1/8 inches Sleeve tab (2 each of hook) 3-1/2 ± 1/8 inches Sleeve hem (2 each of loop) 4-1/2 ± 1/8 inches Front, right (3 each of hook) 2 ± 1/8 inches Front, overlap, left (3 each of loop) 2 ± 1/8 inches						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) Trousers: Hip pocket (4 each of loop) 1 ± 1/8 inches Hip pocket flap (4 each of hook) 1 ± 1/8 inches i. Cut the 5/8 inch tape for front hip adjustment straps 3 ± 1/4 inches for all sizes. Cut the 5/8 inch tape for back hip adjustment straps in the following lengths (± 1/4 inch tolerance). Size — Inches (relaxed) XXX-Small 7-3/4 XX-Small 7-3/4 X-Small 7-3/4 Small 9-1/4 Medium 9-1/4 Large 9-1/4 X-Large 9-1/4 XX-Large 9-1/4 j. Cut two lengths of cotton tape for the chest pocket pulls 4 ± 1/4 inches in length. or k. Cut from outer shell ends for finished measurements of 3/8 to 1/2 inch wide and 4 ± 1/4 inches in length. l. The top sleeve elbow patch, undersleeve elbow patch, atropine pocket, and knee patch shall be cut						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) from one lot of butyl material. The chest pocket protective piece, sleeve pocket protective piece, cargo pocket protective piece, and hip pocket protective pieces may be cut from ends. 2. Replacement of damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages, as classified in 4.4.3 are excluded and replaced with nondefective and properly matched material. 3. Marking. a. The outer shell parts of the suit shall be marked, ticketed or bundled to insure a uniform size and shade throughout the garments. Any method of marking the outer shell parts may be used except: 1. Metal fastening devices. 2. Sew-on type markings. 3. Adhesive type tickets which discolor the material or leave traces of paper or adhesive upon removal of tickets. 4. Punched or drill holes. b. The lining material parts of the suit shall be marked and bundled to insure a uniform size throughout the garments. Each bundle shall have the surveillance number clearly identified. Only one surveillance number can be used in a uniform.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Marking. (cont'd)						
	c. The butyl cut parts shall be bundled after the heavy side is marked to insure, when attached to the areas to be protected, the heavy side shall face toward the outside of the suit.						
4.	Make seven belt loops, two suspender loops and three slide fastener thongs (see operation 1g).						
	a. Fold under side raw edges with edge abutted or overlapped and double stitch length of stripping, with the stitching 3/16 to 5/16 inch gage, using a cover stitch on the underside.	406	EFh-1	6-10	50	50	
	or						
	b. Fold stripping in half lengthwise with the raw edges turned in and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-14	50	50	
	c. The loops and thongs shall finish 3/8 to 1/2 inch in width.						
5.	Attach thongs to slide fastener pulls on trousers (see operation 4).						
	a. Cut the thongs in 6 ± 1/4 inches length.						
	b. Thread one end of thong through end of slide fastener pull and with ends even, secure the ends with a bartack.	Bartack		28 per bartack	50	50	
	or						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN/ LOOPER	THREAD COVER
5.	Attach thongs to slide fastener pulls on trousers (see operation 4). (cont'd)						
	c. Fold length of thong in half. Insert loop end through the end of the pull. Pass the two free ends through the loop and pull up tight.						
	NOTE: Pull thongs are not required on coat slide fastener pull.						
	MAKE COAT						
6.	Make collar.						
	Position outer shell of collar and collar lining face to face and stitch along top and side edges. Turn, work out edges and stitch 1/4 to 5/16 inch from top and side edges.	301 or 401 and 301	SSe-2(a) SSe-2(b)	8-10 8-10	50 50 50	50 70 50	
7.	Make chest pockets.						
	Turn front edges of pocket to inside and place hook portion of nylon fastener tape on inside of pocket, positioned as indicated on pattern. Fold a piece of 3/8 inch cotton tape in half lengthwise (for chest pocket pull), if using material ends for chest pocket pulls construct as specified in operation 4 and place between fastener tape and pocket, at the midpoint of the fastener tape. Stitch 1/8 to 3/16 inch from edges of fastener tape catching the pocket pull on both sides.	301 or 401	LSbj-1	10-14	50 50	50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make chest pockets. (cont'd) NOTE: When the operation is completed, the pocket pull shall be caught securely between the two long rows of stitching attaching the fastener tape to the pocket.						
8.	Attach chest pocket protective piece to outer shell. Position chest pocket protective piece with heavily butyl coated surface facing outward on the finished garment on front outer shell where indicated on pattern. Stitch 1/16 to 1/8 inch from all edges. Place loop portion of nylon fastener over the protective piece, positioned as indicated on front pattern. Stitch 1/8 to 3/16 inch from each edge of tape. NOTE: When using a double needle machine for operations 7 and 8 for attaching fastener tapes, it is only necessary to stitch along the longer sides of the fastener tape.	301	LSbj-1	10-14	50	50	
9.	Attach chest pockets to front outer shell. Position chest pockets over the chest pocket protective piece matching hook and loop portion of nylon fastener. Turn under and stitch the remaining front pocket sides, as well as all other pocket sides 1/16 to 1/8 inch from turned under edges. Place a horizontal bartack at each end of the pocket opening. The protective piece shall not extend beyond the sides and bottom edges. Pocket opening shall be a minimum of 5-7/8 inches.	301 Bartack	LSd-1	10-14 28 per bartack	50 50	50 50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Attach hook fastener tape to outer shell center front edge.						
	Position hook fastener tapes on right coat front (shell only) as indicated on pattern and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	50	50	
11.	Join shoulder seams of outer shell.						
	Join shoulder seams with a double-lapped and double-stitched seam with fronts overlapping backs.	301 or 401	LSc-2	10-14	50 50	50 70	
12.	Labels.						
	Sew identification, size, and instruction label, or combination label 1/8 to 3/16 inch from the edge on on all four sides to the knit side of lining on right front with bottom edges 1-1/4 ± 1/4 inches from bottom edge of front and positioned as follows:	301	LSbj-1 and SSa-1	8-10	50	50	
	(1) Size or combination label - centered (± 1/2 inch tolerance) on right front.						
	(2) Identification label (when used) - positioned to left of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching						
	(3) Instruction label - positioned to the right of size label with nearest side edge abutted. The nearest side edges may be overlapped and caught in one row of stitching.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Labels. (cont'd) NOTE: Final row of stitches on overlapped edges may be continued to bottom of lining.						
13.	Join shoulder seams of lining.						
	Position lining fronts and backs face to face and join shoulder seams. Turn seam allowances to one side and raise stitch 3/16 to 1/4 inch from turned edges.	301 or 401	LSq-2	8-10	50 50	50 70	
14.	Attach collar.						
	a. Position collar to outer shell of coat, face to face, and join along neck edge from notch to notch as indicated on pattern.	301 or 401	SSa-1	10-14	50 50	50 70	
	b. Position lining to outer shell, face to face, with fronts and neck edges even, and join plies along front edges with the stitching continued along entire neck edge including collar.	301 or 401	SSe-2(a)	8-10	50 50	50 70	
	c. Turn coat to right side out, work out edges, and edge stitch fronts 1/4 to 5/16 inch from front edges with the stitching continued across the entire neck edge 1/4 to 5/16 inch from collar seam.	301	SSe-2(b)	8-10	50	50	
	d. When "c" above is completed, the edge of the collar, right and left front, shall finish 2-1/8 ± 1/8 inches from finished front edge.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026/10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Attach elbow patches to outer shell.						
	a. Position the under sleeve elbow patch on the wrong side of the under sleeve and stitch 1/16 to 1/8 inch around edges as indicated on the pattern. In a similar fashion position the top sleeve elbow patch on the wrong side of the top outer shell sleeve as indicated on pattern and stitch 1/16 to 1/8 inch around the edge.	301	LSbj-1	10-14	50	50	
	or						
	b. Stitch under sleeve elbow patch to top sleeve elbow patch. Overlap shall be 1/2 inch.	301	LSa-1	10-14	50	50	
	c. The one piece elbow patch assembly shall be positioned, as indicated on pattern (to wrong side of outershell) after underarm seam is stitched. Stitch 1/16 to 1/8 inch around the edge.	301	LSbj-1	10-14	50	50	
16.	Make sleeve pocket.						
	a. Make one 3/16 to 1/4 inch buttonhole or sewn eyelet at bottom of each sleeve bellows pocket, positioned as indicated on pattern.	401 or		19-24 per eyelet	50	50 or 70	
		502 or		26-32 per eyelet	70	50 or 70	
		503		26-32 per eyelet	70	50 or 70	
	b. Turn top edge of pocket to inside, as indicated on the pattern, and with the raw edge turned in, single stitch 1/16 to 1/8 inch from edge. The hem shall finish 1 ± 1/8 inches wide.	301 or 401	EFb-1	10-14	50 50	50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Make sleeve pocket. (cont'd)						
	c. Position loop portion of nylon fastener, as indicated on pattern, and stitch 1/8 to 3/16 inch from all edges.	301 or 401	LSbj-1	10-14	50	50	
	d. Stitch the cut-out corner of the pocket with a 3/8 inch seam forming the bellows. Turn seam toward the bottom of pocket and raise stitch 1/16 to 1/8 inch from turned edge.	301 or 401 and 301	LSq-2(a) LSq-2(a) LSq-2(b)	10-14 10-14 10-14	50 50 50	50 70 50	
	or						
	e. Stitch corner with two rows of stitching 1/32 to 1/8 inch apart using a 1/2 inch seam.	301 or 401	SSa-2 SSa-2	10-14	50 50	50 70	
	f. As indicated on pattern fold both the side and bottom with the wrong side together and raise stitch 1/8 to 3/16 inch from edges. Fold face to face to form the bellows on the side and bottom and raise stitch 1/8 to 3/16 inch along the side and bottom.	301	OSf-1	10-14	50	50	
17.	Attach sleeve pocket to the outer sleeve.						
	a. Position the pocket protective material as indicated on the pattern and stitch 3/32 -1/16, +0 inch around all the edges.	301	LSbj-1	10-14	50	50	
	b. Position sleeve pocket on the sleeve pocket protective piece, with the bellows towards the rear, and turning in the raw edges stitch along the edges with a 1/16 to 1/8 inch raised stitch. The protective piece shall not extend beyond the sides or bottom of the sleeve pocket.	301	LSd-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Make atropine pocket flap.						
	a. Fold the atropine pocket flap lengthwise and stitch 3/8 ± 1/16 inch seam on each end. Turn the flap working out the corners, and place a row of stitching 1/4 ± 1/16 inch on the closed sides of the flaps, and continue along the folded edge.	301 or 401 and 301	SSe-2(a) SSe-2(b) and OSf-1	10-14 10-14	50 50	50 50	
	b. Position hook portion of the nylon fastener tape on atropine pocket flap as indicated on pattern and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	50	50	
19.	Make atropine pocket and opening.						
	a. Position butyl pocketing on sleeve liner (foam side) and stitch around opening, as indicated on pattern.	301	SSa-1	10-14	50	50	
	b. Tongue notch opening and turn pocketing to inside. Topstitch 1/8 ± 1/16 inch from folded edge. The opening is to finish 3/4 +1/4, -1/16 inch wide.	301	SSa-1	10-14	50	50	
	c. Position pocket facing shell on topsleeve and stitch around opening, as indicated on pattern. Insert pocket flap catching flap in stitching.	301	SSa-1	10-14	50	50	
	d. Tongue notch pocket opening. Turn facing to inside and topstitch 1/8 ± 1/16 inch from all four edges.	301	SSa-1	10-14	50	50	
	e. Superimpose topsleeve pocket opening on top of lining opening. Stitch lower pocket openings together and ends of pocket opening 1/8 ± 1/16 inch from edge.	301	SSa-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Make atropine pocket and opening. (cont'd)						
	f. Close the top and sides of opening by folding down the outershell sleeve and stitching thru the facing and flap raw edges while penetrating sleeve liner and butyl pocketing.	301	SSa-1	10-14	50	50	
	g. Fold butyl pocketing up, so that the top and bottom edges are even, and stitch pocketing to liner 1/8 to 1/4 inch from edge on all four sides. h. When this operation is complete the pocket depth shall finish 6 ± 1/2 inches. NOTE: When this operation is complete and the fastener tapes are engaged, the atropine pocket flap should not lay flat. (The blouse effect is to accommodate the atropine kit.) or	301	SSa-1	10-14	50	50	
	i. Superimpose topsleeve shell (face side up) on topsleeve lining (foam side up). Position atropine flap on shell, position butyl pocket on flap and shell and stitch with 2 rows of stitching 3/4 inch apart thru all plies, catching flap in top row of stitching.	301	SSa-1	10-14	50	50	
	j. Tongue notch as indicated on pattern, turn butyl pocket thru opening to inside and topstitch 1/8 ± 1/16 inch from folded edge of opening on all four sides. Opening is to finish 3/4 +1/4, -1/16 inch wide.	301	SSa-1	10-14	50	50	
	k. Fold butyl pocketing up so that the top and side edges will be even and stitch pocket to lining 1/8 to	301	SSa-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Make atropine pocket and opening. (cont'd) 1/4 inch from edge on all four sides. When this operation is complete, the pocket depth shall finish 6 ± 1/2 inches. NOTE: When this operation is complete, the fastener tapes are engaged, the atropine pocket flap should not lay flat. (The blouse effect is to accommodate the atropine kit).						
20.	Make sleeve tabs.						
	a. Fold under both long raw edges along the seam line and then fold crosswise leaving one raw edge 3/8 ± 1/16 inch shorter than the other. Stitch 1/8 to 3/16 inch from the edge of the tab.	301	SSc-1	10-14	50	50	
	b. Position hook tape on each tab as indicated on the pattern and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	50	50	
21.	Join sleeves.						
	a. Position loop tape on top sleeve shell as indicated on pattern and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	50	50	
	b. Position the raw edge of the sleeve tab even with the raw edge of the top sleeve as indicated on the pattern taking care to keep the hook and loop in alignment. Stitch 1/8 to 3/16 inch from edge.	301 or 401	SSa-1	8-10	50 50	50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Join sleeves. (cont'd)						
	c. Join the top sleeve outer shell to the under sleeve outer shell with a double-lapped, double-stitched seam. The top sleeve shall overlap the under sleeve.	301 or 401	LSc-2	10-14	50 50	50 70	
	d. At back arm, join the top sleeve lining to the under sleeve lining with a 1/2 +1/8, -0 inch seam.	301 or 401	SSa-1	8-10	50 50	50 70	
22.	Set sleeves.						
	a. Join the outer shell sleeve to the outer shell coat with a double-lapped, double-stitched seam. The coat shall overlap the sleeve.	301 or 401	LSc-2 LSc-2	10-14 10-14	50 50	50 70	
	b. Join sleeve lining to coat lining with a 1/2 + 1/8, - 0 inch seam.	301 or 401	SSa-1 SSa-1	8-10 8-10	50 50	50 70	
	c. Position sleeve tab parallel to raw edge of under sleeve hem located where indicated on the pattern such that the raw edges of the sleeve tab and the raw edge of the under sleeve are even.						
	d. In one operation join the outer shell underarm and outer shell side body seams with a double-lapped, double-stitched seam. The front shall overlap the back and the raw edge of the sleeve tab shall be securely caught in the seam.	301 or 401	LSc-2 LSc-2	10-14 10-14	50 50	50 70	
	e. Turn sleeve tab back over the seam and raise stitch 1/2 to 5/8 inch from the turned edge. or	301	SSb-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Set sleeves. (cont'd)						
	f. As an alternate to c., d. and e. above, the sleeve tab may be stitched to the undersleeve, located as indicated on the pattern, however, with the raw edge of the tab 3/8 + 1/8, - 0 inch from the raw edge of the under sleeve.	301 or 401	SSa-1	8-10	50 50	50 70	
	g. After joining the underarm and side seams of the outer shell, as specified in d. above, turn tab back and raise stitch as specified in e. above.						
	h. In one operation, join the lining underarm and side body seams with a 1/2 + 1/8, - 0 inch seam.	301 or 401	SSa-1 SSa-1	8-10 8-10	50 50	50 70	
	NOTE: When completed, the sleeve tabs will fasten from front to back. Cuff adjustments shall be stitched through the shell cloth only.						
23.	Finish sleeve bottoms.						
	With lining even with the finished hem of the outer shell, turn in bottom edge of outer shell as indicated on pattern. Fold in raw edges of outer shell 1/4 ± 1/16 inch and stitch hem completely around the bottom from the turned edge.	301	SSl-1	10-14	50	50	
24.	Finish the bottom of coat.						
	a. Place two metal grommets as indicated on pattern.						
	or						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
24.	Finish the bottom of coat. (cont'd)						
	b. Make two buttonholes or sewn eyelets at the bottom of coat as indicated on pattern in lieu of metal eyelets.	401 or 502 or 503		19-24 per eyelet 26-32 per eyelet 26-32 per eyelet	50 70 70	50 or 70 50 or 70 50 or 70	
	c. Position the elastic drawcord on inside of coat with ends exiting through eyelets and knot ends of drawcord.						
	d. Turn up bottom edge of outer shell even with the lining and with the raw edge turned in stitch 1/8 to 3/16 inch from turned edge with the stitching continuing along the complete bottom. The hem shall finish 1-1/4 ± 1/8 inches from the bottom edge of the coat.	301	SSl-1	10-14	50	50	
	e. Center the elastic drawcord in the hem of the coat. In the center back, place a vertical bartack through the hem catching the waist drawcord.	Bartack		28 per bartack	50	50	
25.	Attach slide fastener.						
	a. Position the back edge of the slide fastener tape on the inside of the left front, 2-1/2 ± 1/8 inches from the finished front edge, as indicated on the pattern, with top stop 1/2 to 5/8 inch from the top.						
	b. Starting at bottom of tape, stitch tape to coat 1/16 to 1/8 inch from back edge, to the top of tape and continue across the turned under end.	301	LSbj-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
25.	Attach slide fastener. (cont'd)						
	c. Place a second row of stitching through tape and coat positioned 1/4 to 5/16 inch from the first row of stitching.	301	LSbj-1	10-14	50	50	
	d. Position the back edge of the slide fastener tape (tape containing slider) on outside of right front 1-7/8 ± 1/8 inches from the finished front edge, with the top stop 1/2 to 5/8 inches from the top.						
	e. Starting at bottom of tape, stitch tape to coat 1/16 to 1/8 inch from back edge to the top of coat and continue across the turned under end.	301	LSd-1 or LSbj-1	10-14	50	50	
	f. Place a second row of stitching through tape and coat positioned 1/4 to 5/16 inch from the first row of stitching.	301	LSd-1 or LSbj-1	10-14	50	50	
	g. When using a double needle machine, turn the coat and continue stitching across the turned under ends of the tape (perpendicular to the slide fastener) and backtack to secure, forming a boxstitch.	301	LSd-2	10-14	50	50	
	or						
	h. Bartack across the turned under ends of the tape (perpendicular to the slide fastener).	Bartack		28 per bartack	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
26.	Attach loop tape fasteners. Position loop tapes on lining side of coat on left center front as indicated on pattern and corresponding to hook tapes on coat right center front. Stitch 1/8 to 3/16 inch from all edges, through all thicknesses.	301	LSbj-1	10-14	50	50	
27.	Attach snap fasteners. Place a female portion snap fastener on the outside of the center back coat bottom, and two additional female snap fasteners, 3 ± 1/8 inches from each side of center back. These fasteners are to be positioned 1-1/4 ± 1/8 inches from bottom of the coat. NOTE: The center snap fastener may be located off center (to either side) enough to allow the male portion to be located just off the trouser back seam.						
	MAKE TROUSERS						
28.	Make fly protective flap.						
	a. Join outer shell and lining piece of protective flap along side and bottom edges, face to face. Turn, work out edges and stitch 3/16 to 1/4 inch from turned edges, with stitching continued to the top edge of outer shell material.	301 or 401	SSe-2(a) SSe-2(b)	10-14 10-14	50 50	50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN/ LOOPER	THREAD COVER
28.	Make fly protective flap. (cont'd)						
	b. Turn in top edge of outer shell even with the raw edge of lining and stitch 1/16 to 1/8 inch from turned edge with the stitching continued along side and top edges of hem. The hem shall finish 1/2 to 5/8 inch in width.	301	SSl-1	10-14	50	50	
29.	Make cargo pocket flap.						
	Fold cargo pocket flap lengthwise and stitch a 1/4 ± 1/16 inch seam on each end. Turn the flap working out the corners and place a row of stitching 1/4 to 5/16 inch on the closed sides of the flap and continue along the folded edge.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b) and OSf-1	10-14 10-14 10-14 10-14	50 50 50 50	50 70 50 50	
30.	Make cargo pockets.						
	a. Make one 3/16 to 1/4 inch sewn eyelet at bottom of each cargo pocket, positioned as indicated on patterns.	401 or 502 or 503		19-24 per eyelet 26-32 per eyelet 26-32 per eyelet	50 70 70	50 or 70 50 or 70 50 or 70	
	b. Fold the two front pleats at marks indicated on pattern and stitch along the fold, 1/8 to 3/16 inch from the edge. NOTE: Pleats shall fold toward the back of pocket as indicated on pattern.	301	OSf-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN/ LOOPER	THREAD COVER
30.	Make cargo pockets. (cont'd)						
	c. Stitch across top and bottom of pleats to hold in position.	301 or 101 or 401	SSa-1 SSa-1 SSa-1	6-14 6-14 6-14	50 50 50	50 70	
	d. With top raw edges of facing and pocket turned under, stitch facing to top inside of pocket, 1/16 to 1/8 inch from edge. NOTE: This operation may be performed with automated folders using facings cut off rolls. No more than one splice per facing.	301 or 401	LSk-2 LSk-2	10-14 10-14	50 50	50 70	
	or						
	e. Stitch the facing to the top of pocket, turn and edge stitch 1/16 to 1/8 inch from top edge.	301 or 401 and 301	SSe-2 SSe-2 SSe-2	10-14 10-14 10-14	50 50 50	50 70 50	
	f. After either of the options, turn in bottom raw edge of facing and stitch to pocket 3/32 to 1/8 inch from the folded edge.	301 or 401	LSk-2 LSk-2	10-14 10-14	50 50	50 70	
	g. Bartack horizontally across fold of pleats. The bartacks shall be not more than 1/8 inch from the lower row of facing stitching.	Bartack		28 per bartack	50	50	
	h. Fold outside edge of bellows and stitch along folded edge (through facing 1/16 to 1/8 inch from folded edge). Place a vertical bartack, 1/8 ± 1/32 inch from top of pocket with bartack superimposed on stitching.	301 and bartack	OSf-1	10-14 28 per bartack	50 50	50 50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026/0-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN/ LOOPER	THREAD COVER
30.	Make cargo pockets. (cont'd)						
	i. Fold bellows pleat at bottom of pocket; fold back side seam allowance in line with folded edge of pleat. Stitch across bottom of pleat catching side seam allowance in the stitching.	301 or 401	LSd-1	6-14	50 50	50 70	
31.	Make hip pocket and hip pocket flap.						
	a. Turn top edge of pocket to inside, as indicated on pattern, and with the raw edge turned under 1/4 inch, single stitch 1/16 to 1/8 inch from edge. The hem shall finish 3/4 ± 1/16 inch in width.	301 or 401	EFb-1	10-14	50	50	
	b. Position the loop portion of the nylon fastener tape to the pocket, as indicated on pattern, and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	50	50	
	c. Fold hip pocket flap in half lengthwise and join side edges. Turn flap, work out edges and stitch 1/4 to 5/16 inch on the closed sides of the flap and continue along the folded edge.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b) and OSf-1	10-14 10-14 10-14	50 50 50	50 70 50	
	d. Position the hook portion of the nylon fastener tape to the pocket flap, as indicated on pattern, and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	50	50	
32.	Attach knee patch to the front outer shell.						
	Position the knee patch on the wrong side of front outer shell as indicated on the pattern and stitch 1/16 to 1/8 inch from all edges.	301	LSbj-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
33.	Join trouser fronts to backs at sides.						
	Join the outer right front shell to outer right back shell with a double-lapped, double-stitched seam, the front to overlap the back. In similar fashion join the left front outer shell to the left back outer shell, the front to overlap the back.	301 or 401	LSc-2	10-14	50 50	50 70	
34.	Attach cargo pocket protective piece to outer shell.						
	Position cargo pocket protective piece as indicated on pattern, and stitch 1/16 to 1/8 inch in from all edges.	301	LSbj-1	10-14	50	50	
35.	Attach cargo pockets and flaps to outer shell.						
	Finished appearance. The bellows side of the cargo pocket shall finish toward the back. Pockets shall be uniformly stitched and properly positioned with the top corners of pocket openings covered by the flap. The edges of the cargo pocket protective piece shall not extend beyond the edges of the bellows pocket and flap.						
	a. Mark position of cargo pockets and flaps, as indicated on the patterns.						
	b. Turn in the seam allowance on the bellows side of pocket. Stitch bellows side of pocket to trousers, positioned as marked, with one row of stitching 1/16 to 1/8 inch from edge. The end of the seam shall catch the turned-under seam allowance at bottom of pocket, closing bellows at bottom.	301	LSd-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
35.	Attach cargo pockets and flaps to outer shell. (cont'd)						
	c. Turn in other side and remaining bottom seam allowance of pocket and stitch to trousers with one row of stitching 1/16 to 1/8 inch from edge to bellows side.	301	LSd-1	10-14	50	50	
	NOTE: Operations 35.b. and 35.c. may be performed in one operation to attach the cargo pocket. When using the single operation, the back bottom corner of each cargo pocket shall be backtacked through all layers for not less than 1/4 inch, and continue to stitch up the back side, pivoting at the corner such that the crossover stitch is immediately after the backtack to preclude any weak spot in this area.						
	d. Bartack back and front edges of pocket opening with vertical bartacks in line with pocket stitching through pocket and trousers.	Bartack		28 per bartack	50	50	
	e. Position raw edges of cargo pocket flap above pocket opening as indicated on pattern, and stitch 3/16 to 1/4 inch from raw edges. Turn flap down and raise stitch 1/4 to 5/16 inch from turned edge, burying the raw edges within the seam.	301	LSbl-2	10-14	50	50	
	or						
	f. Turn in top raw edge of flap, positioned on trousers as indicated on pattern, and double-stitch burying raw edges within seam.	301	LSd-2	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026/0-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
35.	Attach cargo pockets and flaps to outer shell. (cont'd)						
	g. Bartack lower front corner of pocket flap with a vertical bartack 1/16 to 1/8 inch from side and bottom edges through pocket flap, pocket and trousers.	Bartack		28 per bartack	50	50	
36.	Attach hip pocket protective piece.						
	Position hip pocket protective pieces as indicated on pattern and stitch 1/16 to 1/8 from all edges to back outer shell.	301	LSbj-1	10-14	50	50	
37.	Attach hip pockets and flaps to outer shell.						
	a. Position hip pocket as indicated on pattern, turn in sides and bottom edges of hip pocket and stitch 1/16 to 1/8 inch from the edges. The protective piece shall not extend beyond side and bottom edges of pocket.	301	LSd-1	10-14	50	50	
	b. Position raw edges of hip pocket flap above pocket opening as indicated on patterns and stitch 3/16 to 1/4 inch from raw edges. Turn flap down and raise stitch 1/4 to 5/16 inch from the turned edge, burying the raw edges within the seam. The protective piece shall not extend beyond top and side edges of flap. The side edges of flap shall align with the side edges of the pocket within ± 1/8 inch.	301	LSbl-2	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
38.	Make slide fastener cover for the leg adjustment.						
	a. As indicated on pattern, join slide fastener cover, face to face, with 1/2 ± 1/32 inch seam for 2 ± 1/8 inches from top.	301 or 401	SSa-1	10-14	50 50	50 70	
	b. Position slide fastener below seam and to inside of each opening with raw edges turned under, and with top slide stop 2-3/4 ± 1/8 inches from open end of slide fastener cover. Turn under slide fastener tapes at top slide stop and stitch slide fastener cover 3/8 ± 1/32 inch from opening with stitching extending around both sides and top forming a U-shape.	301 or 401	LSd-1	10-14	50 50	50 70	
39.	Attach slide fastener and cover to outer seam of outer shell pant leg.						
	Place slide fastener and cover in an open position on the outer shell, with the functional side of the assembly located at the markings indicated on the pattern. Turn in edges and raise stitch 1/16 to 1/8 inch around the edges. NOTE: When slide fastener is closed the leg opening will be narrower and the slider shall be toward the bottom of the trouser leg.	301 or 401	LSd-1	10-14	50 50	50 70	
40.	Join crotch, seat and inseams of outer shell.						
	a. Position left front to right front at crotch and join the crotch seam beginning at the inseam and continuing for 2-7/8 ± 1/8 inches, stopping at notch as indicated on pattern. The seam allowance may be folded to either side. The top ply shall be turned	301 or 401	LSq-3	10-14	50 50	50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
40.	Join crotch, seat and inseams of outer shell. (cont'd)						
	sharply back over the first row of stitching, showing the face of the shell fabric. Two rows of stitching shall be placed at 1/8 and 1/4 inches from crotch seam on side of the seam allowance.						
	or						
	b. Stitch with a double needle machine.	301 or 401	LSc-2	10-14	50 50	50 70	
	c. Join seat seam of outer shell with double-lapped and double-stitched seam. The seat seam may lap either direction.	301 or 401	LSc-2	10-14	50 50	50 70	
	d. Join inseams with a double-lapped and double-stitched seam and the front will overlap the back.	301 or 401	LSc-2	10-14	50 50	50 70	
41.	Make trouser lining (laminate material).						
	Join side seams, crotch seam to notch, seat seam and inseams of trouser lining, stitch all seams with material face to face (knit side to knit side).	301 or 401	SSa-1	8-10	50 50	50 70	
42.	Attach label.						
	Sew identification, size, (combination identification, size, and instruction label, when used) and instruction label 1/8 to 3/16 inch from edge on all four sides to the knit side of left back of lining material (positioned as indicated below) with the top edges 3-1/8 ± 1/8 inches below top edge.	301	LSbj-1	8-10	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
42.	Attach label. (cont'd) (1) Size or combination label - centered (± 1/2 inch tolerance) on left back. (2) Identification label (when used) - positioned to left of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching. (3) Instruction label - positioned to the right of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching. NOTE: Final row of stitches on overlapped edges may be continued to top of lining.						
43.	Join outer shell and lining at front opening. Insert lining into the outershell, turn raw edges under 3/8 inch on outershell and lining between plies at front opening and stitch 1/16 to 1/8 inch from folded edge. The stitching shall continue to the top edge of the outershell and shall terminate at the notch at base of fly.	301	SSc-1	10-14	50	50	
44.	Attach slide fastener to fly opening. a. Position slide fastener to inside or outside of right front with back edge of chain extending beyond front edge 3/16 ± 1/16 inch. Stitch to front with two rows of stitching. The first row shall be 1/16 to 1/8 inch from back edge of tape and the second	301	SSbd-2	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
44.	Attach slide fastener to fly opening. (cont'd)						
	row 1/8 to 3/16 inch from the first row. The top stop shall finish 1-3/8 ± 1/8 inches from top edge of finished waist. If slide fastener is placed on outside the right front; the top end of fastener tape shall be turned under and caught in the stitching. If slide fastener is placed on inside of right front; the stitching shall continue to top end of fastener tape or the tape may be turned under. The stitching shall continue to the bottom end of the slide fastener tape.						
	b. Position slide fastener to inside of left front with edge of opened chain positioned as indicated on pattern and stitch through tape to front of trousers with two rows of stitching. The first row shall be 1/16 to 1/8 inch from back edge of tape and the second row 1/8 to 3/16 inch from the first row. The top end of tape may be turned under. The top stop shall finish 1-3/8 ± 1/8 inches from top edge of finished waist. The stitching shall continue to the bottom end of the slide fastener tape.	301	LSbj-2	10-14	50	50	
45.	Finish waist.						
	Turn in top edge of outer shell one inch as indicated on pattern (even with the lining material) and with the raw edge turned in 1/4 inch, stitch 1/16 to 1/8 inch from turned edge, with the stitching continued along ends of the front opening. The top ends of the slide fastener tapes may be caught in the waist stitching (when applicable). The finished waist hem shall finish 3/4 ± 1/8 inch.	301	SSl-1	10-14	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
46.	Finish leg bottoms.						
	a. Make two horizontal buttonholes or sewn eyelets at the bottom of each leg of the outer shell as indicated on pattern.	401 or		19-24 per eyelet	50	50 or 70	
		502 or		26-32 per eyelet	70	50 or 70	
	or	503		26-32 per eyelet	70	50 or 70	
	b. Secure two grommets in same positions in lieu of buttonholes.						
	c. Position the leg bottom drawcords on inside of leg with end exiting through buttonholes, eyelets or grommets, knot ends of drawcord.						
	d. Turn up bottom of outer shell in accordance with pattern (even with lining material) and with raw edge and slide fastener cover turned in 1/4 inch, stitch 1/8 to 3/16 inch from turned edge, around entire hem and gusset sewing across slide fastener cover.	301	SSl-1	10-14	50	50	
	NOTE: When the hem is stitched, no stitching will interfere with the functioning of the slide fastener or the drawcord.						
	e. Center drawcord in leg bottom hem. Place a vertical bartack through inseam catching leg drawcord.	Bartack		28 per bartack	50	50	
47.	Make and attach adjustment straps.						
	a. Thread front hip adjustment strap over first bar (opposite pull tab of buckle) and fold strap in half lengthwise so that bar is located at the fold. Fold	301		8-10	50	50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
47.	Make and attach adjustment straps. (cont'd)						
	under two raw ends of strap 1/4 ± 1/16 inch. Position strap with long edge parallel to waist and with turned under edges 1-1/4 ± 1/16 inches from outseam on the front, and 3/4 ± 1/8 inch below finished waist edge. Stitch to trousers with a 3/8 to 1/2 inch box-x stitch pattern, 1/16 to 1/8 inch from turned under edges.						
	or						
	b. Make two bartacks across the width of the strap, the first being 1/16 to 1/8 inch from the turned under edge and the second 1/8 to 1/4 inch away from the first bartack.	Bartack		28 per bartack	50	50	
	c. Turn back one end of back hip adjustment strap 1 ± 1/8 inch and then turn under raw end 1/4 ± 1/16 inch and boxstitch 1/16 to 1/8 inch from edge.	301		8-10	50	50	
	or						
	d. Bartack 1/8 ± 1/16 inch from turned under edge.	Bartack		28 per bartack	50	50	
	e. Thread unfinished end of back hip adjustment strap through pull tab end of buckle by bringing the strap up from underneath the buckle through the pull tab and the first bar, over the first bar and down through the buckle again. Pull unfinished end of strap toward trouser back until turned under end is at the pull tab.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
47.	Make and attach adjustment straps. (cont'd)						
	f. Turn under unfinished end of back hip adjustment strap 1/2 ± 1/8 inch. With strap fully extended laying flat, and 3/4 ± 1/8 inch below finished waist edge. Stitch to trousers with a 3/8 to 1/2 inch box-x stitch pattern, 1/16 to 1/8 inch from turned edges.	301		8-10	50	50	
	or						
	g. Make two bartacks across the width of the strap, the first being 1/16 to 1/8 inch from the turned under edge and the second 1/8 to 1/4 inch away from the first bartack.	Bartack		28 per bartack	50	50	
48.	Attach belt loops.						
	a. Except for sizes XXX-Small and XX-Small, all other sizes shall have seven belt loops. Sizes XXX-Small and XX-Small shall have five belt loops. All belt loops shall finish 1-5/8 ± 1/8 inches in length (measurement taken between bartacks) on each pair of trousers positioned as follows:						
	(1) Except for sizes XXX-Small and XX-Small, all other sizes shall have one belt loop on each front with the front edges of loops 2-5/8 ± 1/8 inches from respective finished edges of front opening.						
	(2) Except for sizes XXX-Small and XX-Small, all other sizes shall have one belt loop on each front 1/4 to 3/8 inch in front of the adjustment strap turn-under.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
48.	Attach belt loops. (cont'd)						
	(3) For sizes XXX-Small and XX-Small, one belt loop centered ± 1/4 inch on each front from respective finished edges.						
	(4) One centered on each back part.						
	(5) One adjacent to seat seam.						
	b. Turn under raw edge of belt loop and position even with the top edge of waist. Bartack to trousers 1/8 to 3/16 inch from top of folded edge.	Bartack		28 per bartack	50	50	
	or						
	c. Position raw edge of belt loop 1/4 inch from top edge of finished waist. Bartack to trousers 1/8 to 3/16 inch from edge of waist. Fold loop over and then proceed as specified in d.	Bartack		28 per bartack	50	50	
	d. Turn under bottom raw edge to finished position and bartack 1/8 to 3/16 inch from folded edge.	Bartack		28 per bartack	50	50	
49.	Attach the fly protective flap.						
	a. Position protective flap on inside of left front, as indicated on pattern, with the outer shell toward the outside and join to trousers and stitch 3/16 to 1/4 inch from back and bottom edge of flap. The fly protective flap may be bartacked at crotch.	301 or Bartack	LSbj-1	10-14 28 per bartack	50 50	50 50	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
49.	Attach the fly protective flap. (cont'd)						
	b. Place a vertical bartack at top back edge of flap through trousers. The bartack shall be positioned 1/4 to 5/16 inch from top and back edges of flap.	Bartack		28 per bartack	50	50	
	or						
	c. As an alternate the bartack may be placed even with the top edge of fly protective flap and 1/4 to 5/16 inch from the back of the flap.						
50.	Attach suspender loops.						
	a. Cut suspender loops 6 ± 1/4 inches long.						
	b. Position a loop on each front with the back end of loop 3/8 ± 1/8 inch in front of each side seam. The other end of the loop shall extend 4-1/4 ± 1/4 inches toward the front of the trousers. Both ends of each loop shall be attached 1/4 ± 1/16 inch below the top of finished edge of waist as follows:						
	(1) Turn under end 1/2 ± 1/16 inch.						
	(2) Set end at 45 ± 10 degree angle.						
	(3) Bartack 1/8 to 3/16 inch from turned under end across width of the loop.	Bartack		28 per bartack	50	50	
	c. Finished appearance shall be as shown in figure 2.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026/10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
51.	Attach snap fasteners. a. Secure a double stud snap fastener to right front at waist, or as an alternate secure style 2 or 2A stud-eyelet combination (on right outside of waist opening) and socket (on right side of waist opening) as indicated on pattern. b. Secure a female portion of snap fastener to left front at waist, positioned to correspond with male at right front at waist. c. Secure a socket or as an alternate a stud (if alternate in 51.a. is used) portion of snap fastener to fly protective flap, positioned to correspond with the double stud fastener on the right front. The fly protective flap shall lay smooth and flat when the socket and stud are engaged. d. Secure two female portions of snap fasteners to each cargo pocket flap, as indicated on pattern. e. Secure two male portions of snap fasteners to each cargo pocket to correspond to female portions on pocket flap. f. Secure three male portions of snap fasteners to trouser back to correspond to female portions on coat back position as follows: (1) Position the male portion of the snap fastener 2-1/4 ± 1/8 inches from finished waist edge for sizes medium, large, X-large and XX-large.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026/10-21T23:01Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
51.	Attach snap fasteners. (cont'd) (2) Position the male portion of the snap fastener 1-3/4 ± 1/8 inches from the finished waist edge for sizes XXX-small, XX-small, X-small and small.						
52.	Clean coat and trousers. a. Trim end of threads to 1/4 inch maximum length. b. Remove all loose threads. c. Remove all size markings.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

3.9 Finished measurements. The finished measurements for coat and trousers shall be as shown in table II.

TABLE II. Finished measurements (inches)

	COAT		TROUSERS	
	1/2 Chest 1/	Back Length 2/	1/2 Waist 3/	Inseam 4/
XXX-Small	17	28	13	32
XX-Small	19	28-1/2	15	32
X-Small	21	29	17	32
Small	23	29-1/2	19	34
Medium	25	30	21	34
Large	27	30-1/2	23	36
X-Large	29	31	25	36
XX-Large	31	31-1/2	27	36
Tolerance	± 3/4	± 3/4	± 3/4	± 1/2

1/ Measure across chest from folded side edge to folded side edge at base of armholes.

2/ Measure down center of back from collar joining seam to bottom edge of coat.

3/ Measure across waist from side edge to side edge with snap fastener at waist secured and fly closed.

4/ Measure from crotch seam along inseam to bottom edge of leg.

3.10 Carbon tetrachloride absorption. When tested in accordance with 4.4.5, the carbon tetrachloride absorption of the laminate material in the end item shall show no individual sample result less than 1.3 mg/cm^2.

3.11 Workmanship. The finished suit shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Utmost care shall be taken during component and subassembly fabrication to ensure quality workmanship and safety of the service person using the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4, and tested as required in table IV.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

4.4.1. Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Toxicological agents resistance testing. Testing for conformance to initial toxicological agents resistance, shall be in accordance with section 4 of MIL-C-12189, except that after 25 consecutively produced rolls have successfully met the requirements, one out of every two rolls shall be randomly sampled and tested. In the event of a failure under the reduced sampling, 100 percent sampling shall be reinstated until an additional 25 consecutive rolls have successfully met the requirements.

4.4.2 In-process inspection. In-process inspection shall be performed on the cloth and foam sides of laminated lining material during the cutting operations and fabrication of the lining. Any cut, tear, needle chew, hole, burn (not repaired in accordance with 3.6.5), or mend (not repaired in accordance with 3.6.5), shall be cause for rejection of the lining. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance (see 6.5.2).

4.4.2.1 Critical defect examination of end items. Prior to performing the end item sampling examination required in 4.4.3, the coats and trousers shall be 100 percent examined for the critical defects in table III. This 100 percent critical defect examination shall become a part of the contractor's inspection system or quality program.

4.4.2.2 Desiccant exposure to atmosphere. The contractor's procedure for ensuring that desiccant units are not exposed to the atmosphere exceeding 2 hours (see 5.1.1) from time of unsealing component packaging to sealing in unit packs shall be described in his/her quality control or inspection plan.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of suits. The sample unit shall be one suit. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total (major and minor combined) defects. The AQLs shall be applied separately to the coats and trousers and failure of either the coats or trousers shall be cause for rejection of the lot. The finding of one or more critical defects shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS			
Material defects and damages	Applicable to laminated cloth, foam side:			
	a. Any cut, tear, needle chew, hole, or burn thru the laminated cloth:			
	- 1/16 to 1 inch long inclusive		101	
	- more than 1 inch long	1		
	b. Any cut, tear, hole, or burn in knit base fabric:			
	- 1/2 to 1 inch long inclusive			201
	- more than 1 inch long		102	
	c. Any mend:			
	- more than 1-1/4 inches in longest direction		103	
	- more than two mends per garment		104	
	Applicable to cotton/nylon outer shell cloth:			
	a. Any hole, needle chew, cut, tear, or burn		105	
	b. Mends:			
	- any mend up to 1/4 inch in diameter or length			202
	- any mend more than 1/4 inch in diameter or length		106	
	- more than two mends		107	
	c. Any defect such as a smash, or multiple float			203
	d. Any misweave, area of poor dye penetration dye streak, broken or missing yarn, thin place, or shade bar			204
	Applicable to butyl cloth:			
	- Any hole, cut, tear, mend or needle holes in atropine pocket	2		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Material defects and damages (cont'd)	- Any hole, cut, tear, mend or needle holes in top sleeve elbow patch, under-sleeve elbow patch, and knee patch		108	
	- Any hole, cut, tear, mend or needle holes in chest pocket protective piece, and hip pocket protective piece:			
	- up to and including 1-1/4 inch			205
	- over 1-1/4 inch		109	
	- Heavily coated side not facing outward			206
	Applicable to hook and loop fastener tape:			
	- any cut, hole, or tear		110	
	- any missing yarn		111	
	- any broken yarn			207
	- hook or loop flattened		112	
	- width not as specified			208
	- any spot, stain, or streak			209
	- color not as specified			210
Shaded parts	Any outside part shaded except those cut from ends			211
Cleanness	Spot or stain (grease, oil, ink)			212
	NOTE: Stains attributed to charcoal content of lining should not be scored.			
	Five or more thread ends in excess of 1/2 inch, not trimmed from coat			213
	Two or more shade or size tickets or loose threads not removed from coat			214

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Cleanness (cont'd)	Five or more thread ends in excess of 1/2 inch, not trimmed from trousers			215
	Two or more shade or size tickets or loose threads not removed from trousers			216
Components and assembly	Any component part or required operation omitted, unless otherwise specified		113	
	Any assembly or operation not properly performed, unless otherwise classified herein			217
	Any component not as specified		114	
	The size of the coat not matching the size of the trousers		115	
Cutting	Any component part not cut in accordance with directional lines indicated on pattern or not in accordance with specified requirements		116	
Seams and stitching	Accuracy of seaming:			
	a. Seams twisted, puckered, or pleated (unless otherwise classified herein)			218
	b. Part of garment caught in unrelated operation or stitching		117	
	c. Ends of seams or stitching produced with 301 stitch type, when not caught in other seams or stitchings, backtacked less than 1/4 or more than 1/2 inch			219
	d. Stitching overlapping end(s) of thread breaks less than 1/2 or more than 3/4 inch			220

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Seams and stitching (cont'd)	e. Various shades of thread used on outside of same coat or trousers			221
	f. Ends of a continuous line of stitching not overlapped, or overlapped less than 1/2 or more than 3/4 inch			222
	Gage/margin of stitching:			
	a. Gage of double needle stitching more or less than specified		118	
	b. Margin of edge or raised stitching more or less than specified:			
	- up to a length of 2 inches inclusive			223
	- beyond 2 inches in length		119	
	Open seam:			
	a. Any on joining seam of laminate cloth except those listed in c. below:			
	- up to a length of 1/4 inch inclusive			224
	- over 1/4 but not greater than 7/8 inch in length inclusive		120	
	- more than 7/8 inch in length	3		
	b. Any seam except joining seam of laminate cloth and those listed in c. below:			
	- up to a length of 1/2 inch inclusive			225
	- more than 1/2 inch in length		121	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Seams and stitching (cont'd)	c. Any on coat bottom hem, outside collar, trouser waistband turnunder, trouser leg bottom hem, sleeve hem, pocket stitchings, slide fastener tape, or on fastener tape:			
	- 1/4 to 1 inch in length inclusive			226
	- more than 1 inch		122	
	NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches on joining seam constitute an open seam. On double-stitched seams, a seam is considered open when one or both sides of the seam is open.			
	Raw edges (outer shell):			
	a. On outside:			
	- up to a length of 1/2 inch inclusive			227
	- more than 1/2 inch in length		123	
	b. On inside:			
	- any raw edge more than 1/2 inch in length (where edge is required to be turned in)			228
	NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.			
	Runoffs:			
	a. Joining seams: When resulting in an open seam, use "open seam" classification.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Seams and stitching (cont'd)	b. Edge or raisesd stitching (when not resulting in any open seam):			
	- 1/2 to 1 inch in length inclusive			229
	- more than 1 inch		124	
	Seam and stitch type:			
	a. Wrong seam or stitch type			230
	b. Looper thread of 401 stitch type finishing on outside of coat or trousers		125	
	Stitch tension:			
	a. Loose tension, up to 1 inch (excluding laminate lining cloth)			231
	b. Loose tension, more than 1 inch (excluding laminate lining cloth)		126	
	c. Loose tension, up to 1 inch on laminate lining cloth		127	
	d. Loose tension, more than 1 inch on laminate lining cloth	4		
	e. Loose tension on edge of raised stitching			232
	f. Tight tension (stitches break when normal strain is applied to the seam or stitching)		128	
	NOTE: Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Seams and stitching (cont'd)	Stitches per inch (to be scored only when the condition exists on major portion of seam)			
	a. Less than specified:			
	- one or two stitches			233
	- more than two stitches		129	
	b. One or more stitches more than specified			234
	Stitches skipped or broken in any location other than specified in defect c page 58:			
	- up to a length of 1/2 inch inclusive			235
	- over 1/2 inch but not greater than 1 inch in length		130	
	- more than 1 inch in length	5		
Buttonholes or eyelets	Not as specified			236
	Not in specified position			237
Grommets	Not as specified			238
	Not type, class, or size specified			239
	Omitted, damaged, or malformed			240
	Improperly clinched		131	
Slide fastener	Shade of tape not as specified			241
	Any part of slide fastener assembly omitted, bent, broken, cracked, or otherwise defective, affecting function	6		
	Bottom closed-end stop not attached as specified		132	
	Not specified type or size		133	
	Fastener tape cut or torn		134	
	NOTE: The slide fasteners shall be fully closed and opened to determine if the slide fasteners are operable and provide a secure closure.			

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT AND TROUSERS (cont'd)			
Snap fastener	Any omitted, mismated, bent, or broken (except on cargo pockets)		135	
	Any fastener not functioning properly i.e. fails to snap closed, provide a secure closure, or open freely (except on cargo pockets)		136	
	NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.			
	Clinched excessively tight, cutting adjacent material		137	
	Clinched loosely, permitting any component to rotate freely but not to the degree that any component can be expected to become detached during use			242
	Clinched loosely to the degree that components can be expected to become detached during use		138	
	NOTE: Incomplete roll of end of button or eyelet barrel is evidence of improper and insecure clinching.			
	Incorrect style		139	
	More than three splits in eyelet or button barrel			243
	Not located on back of coat as specified			244
	Not aligned with each other creating bulge or twist when closed			245

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT			
Elbow patches	Top sleeve elbow patch missing		140	
	Under sleeve elbow patch missing		141	
	Top sleeve or under sleeve elbow patch not stitched as specified to the back side of the outer shell material		142	
	Out of position by more than 1/4 inch			246
Pockets and flaps	Chest and sleeve pocket protective pieces exposed beyond side or bottom edges of pocket, or exposed beyond side or top edges of flap			247
	Chest pocket pull tape missing or not properly caught in stitching			248
	Edges of pockets pleated or twisted in stitching			249
	Raw edges of pocket hems not turned in			250
	Pocket flap tight causing fullness, twisting or curling of pocket flap			251
	Bellows portion of sleeve pocket positioned toward front of pocket		143	
	Sleeve pocket set on crookedly or poorly shaped			252
	Sleeve pockets out of alignment with each other by more than 1/4 inch			253
	Sleeve pocket not parallel with atropine pocket by 3/8 inch or more			254
	Flap not covering front or back edge of pocket by 3/16 inch or more			255

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT (cont'd)			
Pockets and flaps (cont'd)	Hook and loop portions of fastener tape out of alignment when lying flat:			
	- more than 1/4 inch horizontally			256
	- more than 1/4 inch vertically			257
	Cut lengths of tape for atropine pocket and flaps less than 1-3/8 or more than 1-5/8 inches long			258
	Cut lengths of tape for chest pockets less than 6-3/8 or more than 6-5/8 inches long			259
	Cut lengths of loop tape for sleeve adjustments less than 4-3/8 or more than 4-5/8 inches			260
	Cut lengths of hook tape for sleeve adjustment tab less than 3-3/8 or more than 3-5/8 inches			261
	Cut lengths of tape for center front overlap less than 1-7/8 or more than 2-1/8 inches in length			262
	Cut length of pull tape for chest pockets less than 3-3/4 or more than 4-1/4 inches			263
	Bartack stitching loose, incomplete, broken, not fully engaging stitched material, or not as specified:			
	- involving one bartack			264
	- involving two or more bartacks		144	
	Chest pockets out of parallel with respect to finished center front edge by more than 1/4 inch			265

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:01Z
Check the source to verify that this is the currentversion before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT (cont'd)			
Pockets and flaps (cont'd)	Chest pockets out of alignment between corresponding points by more than 1/2 inch (determination to be made by measuring from bottom of coat)			266
Slide fastener	Top ends of tape not turned in and caught in the stitching joining slide fastener to coat			267
	Top stop less than 1/2 inch or more than 5/8 inch from neck edge, or bottom edge of tape more than 1/4 inch from bottom of coat or extending beyond bottom edge			268
	Back edge of slide fastener tape less than 1-3/4 or more than 2 inches from right front edge			269
	Back edge of slide fastener tape less than 2-3/8 or more than 2-5/8 inches from front left coat front edge			270
Collar	Edges of collar ends out of alignment by more than 1/4 inch (place shoulder seams together and extend collar ends to determine alignment)			271
Sleeves	Top sleeve not lapping under sleeve			272
	Ends of underarm seam and side seam staggered more than 1/2 inch (measured from center to center) of double rows of stitching			273
	Sleeve hem less than 1 inch or more than 1-1/4 inches in width			274
	Sleeve tab set on crookedly or poorly shaped			275

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT (cont'd)			
Sleeves (cont'd)	Sleeve tab and loop tape out of alignment (measured at point of attachment to finished hem bottom in parallel direction):			
	- more than 3/8 and less than 3/4 inch			276
	- 3/4 inch or more		145	
	Sleeve tab and loop tape not able to be fastened	7		
	Width of sleeve tab less than 1 inch and more than 1-3/16 inch			277
Shoulder and armhole seams of outer shell	Fronts and back not lapping sleeves at armhole			278
	Fronts not lapping back at shoulder			279
Coat lining	Knit side of laminate material not finishing toward the inside	8		
	Seam allowance of joining seams not finishing toward the outer shell		146	
Bottom hem	Inside edge of eyelet end of buttonhole less than 4 or more than 4-1/2 inches from finished edge of respective front			280
	Elastic drawcord caught in the stitching			281
	Hem less than 1-1/8 or more than 1-3/8 inches in width			282
Labels	Labels not positioned and attached to the knit side of right front lining as specified			283

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	COAT (cont'd)			
Labels (cont'd)	Size, instruction, identification, or combination label (when applicable) missing, incorrect, or illegible			284
	Surveillance marking omitted, incorrect or illegible		147	
	Label not stitched on all four sides			285
	Stitched through printed portion of label			286
	TROUSERS			
Leg slide fastener cover	Leg slide fastener covers stitched through lining material		148	
	Edges of leg slide fastener covers not abutted or overlapping 1/16 inch or more when fastener is closed			287
	Ends of zipper tape not caught in the line of stitching at top of leg opening			288
	With slide fastener in closed position, slider at top of opening	9		
	Thong omitted on trousers slide fastener pull			289
	Slide fastener top stop (at trouser hem) located more than 1-1/4 inches from finished hem			290
Suspender loops	End of loop not turned under or not attached to inside of trouser on waist hem			291
	End of loop not below finished edge of waist 3/16 to 5/16 inch			292

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	TROUSERS (cont'd)			
Suspender loops (cont'd)	Back end of loops not located 1/4 to 1/2 inch in front of side seams			293
	Loop finishing less than 4 or more than 4-1/2 inches in length between center of bartacks			294
	Bartack not at angle specified			295
	Loop less than 3/8 or more than 1/2 inch in width			296
	Bartack stitching loose, incomplete, broken, not fully engaging stitched material, or not as specified:			
	- involving one bartack			297
	- involving two or more bartacks		149	
	Suspender loop missing		150	
Belt loops	Less than 3/8 or more than 1/2 inch wide			298
	Less than 1-1/2 or more than 1-3/4 inches in length between bartacks			299
	Front edges of belt loops on fronts less than 2-1/2 or more than 2-3/4 inches from respective finished edges of front opening (not applicable to sizes XX-Small and XXX-Small)			201a
	One or both side loops not on front or more than 1-1/4 inches from outseam			202a
	Back loops more than 1/2 inch off center between outseam and seat seam			203a
	Center back loop not positioned adjacent to seat seam			204a
	Set with the covering stitching on the outside			205a

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	TROUSERS (cont'd)			
Belt loops (cont'd)	Top folded edge extending above or below top edge of waist more than 1/8 inch			206a
	Bartack stitch loose, incomplete, broken, not fully engaging stitched material, or not as specified:			
	- involving one bartack			207a
	- involving two or more bartacks		151	
	One missing			208a
	Two or more missing		152	
	XX-Small and XXX-Small belt loops not centered on front			209a
Hip pockets and flaps	Hip pocket protective piece exposed beyond side or bottom edges of pocket, or exposed beyond side or top edges of flap			210a
	Hook and loop portions of fasteners out of alignment when flap is lying flat:			
	- more than 1/4 inch horizontally			211a
	- more than 1/8 inch vertically			212a
	Cut lengths of tape less than 7/8 or more than 1-1/8 inches			213a
	Pockets and pocket flaps out of alignment with each other by more than 1/4 inch			214a
	Pockets and pocket flaps out of alignment at any corresponding point, by more than 1/2 inch when measured from top edge of finished waist			215a

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	TROUSERS (cont'd)			
Cargo pockets and flaps	Pleats turned toward front			216a
	Bellows side finished on front			217a
	Corner of flap caught in bartacks at end of bellows opening, restricting freedom of flap			218a
	Pocket flap not uniform in shape and size by more than 1/4 inch			219a
	Pocket flap not completely covering pocket opening by 3/16 inch or more			220a
	Snaps omitted or improperly placed			221a
	Raw edge of pocket hems not turned in			222a
	Pocket flap tight causing fullness, twisting, or curling of pocket flap			223a
	Edge of pockets pleated or twisted in stitching			224a
	Pocket and pocket flaps out of alignment with each other by more than 1/4 inch			225a
	Cargo pocket protective piece exposed beyond side or bottom edges of pocket, or exposed beyond side or top edges of flap			226a
	Top of pockets or pocket flaps out of alignment, at any corresponding point by more than 1/2 inch when measured from top edge of finished waist			227a
Snap fasteners	Not located as specified			228a
	Not aligned with each other creating excessive bulge or twist when closed			229a

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	TROUSERS (cont'd)			
Fly protective flap	Bartack not as specified			230a
	Bartack not located as specified			231a
Hip adjustment strap and buckle	Improperly threaded, i.e., causing buckle to finish on reverse side or to function improperly		153	
	Top end of strap positioned more than 7/8 or less than 5/8 inch from top of finished waist			232a
	Turned under edge of front hip adjustment strap less than 1-3/16 or more than 1-5/16 inches forward of the outseam			233a
	Hem on free end of adjustment strap less than 5/8 or more than 3/4 inch or not box-stitched or bartacked as specified			234a
	Buckle defective or burred, affecting function		154	
	Adjustment straps (when loosened) not extending full length on back of trousers, or extending beyond their length, i.e., not permitting full expansion of top of trousers			235a
Front opening	Top stop on fastener tape less than 1-1/4 or more than 1-1/2 inches from finished top edge of waist			236a
	Back edge of slide fastener chain extending less than 1/8 or more than 1/4 inch beyond right front edge of opening			237a

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T23:01Z
Check the source to verify that this is the currentversion before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	TROUSERS (cont'd)			
Front opening (cont'd)	Top edge of slide fastener tape not turned under when positioned on outside of right front			238a
	Double stud snap fastener not secured to right front			239a
	Thong omitted on trouser slide fastener pull			240a
Knee patches	Knee patch missing		155	
	Knee patch not stitched as specified to the back side of the outer shell material		156	
	Out of position by more than 1/4 inch			241a
Drawcords	Dipping or impregnation omitted on any end			242a
	Knot omitted on both ends of one or more cords			243a
	Dipping or impregnation less than 1/2 inch in length			244a
	Any drawcord less than 29 or more than 31 inches			245a
Finished waist	Top hem turned down less than 5/8 or more than 7/8 inch			246a
Outseams, inseams, and crotch seam	Fronts not lapping back			247a
	Left front not lapping right front		157	
	Crotch and seat seam staggered more than 1/4 inch			248a
	Opening in crotch area, not covered by protective flap	10		
Leg hems	Eyelet end of buttonhole not toward inseam			249a
	Drawcord caught in hem stitching			250a
	Bartack at inseam missing or not catching drawcord			251a

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
	TROUSERS (cont'd)			
Leg hems (cont'd)	Bottom hem turned up less than 7/8 or more than 1-1/8 inches			252a
	Finished bottom hem less than 5/8 or more than 7/8 inch			253a
Crotch seam (joining outer shell to lining)	Stitching joining outer shell to lining extending less than 1-1/4 or more than 1-1/2 inches below fly opening, or extending less than 3/4 or more than 1 inch above fly opening			254a
	Stitching joining outer shell to lining extending less than 3/16 or more than 5/16 inches across fly, or not superimposed on inside row of crotch joining seam			255a
Trousers lining	Knit side of laminate material not finishing toward the inside	11		
	Seam allowances of joining seams not finishing toward the outer shell		158	
Labels	Label not positioned and attached to the knit side of left back lining, as specified			256a
	Size label, instruction label, identification label, or combination label (when applicable) missing, incorrect, or illegible			257a
	Surveillance marking omitted, incorrect, or illegible		159	
	Label not stitched on all four sides			258a
	Stitched through printed portion of label			259a

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

4.4.4 End item examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of suits. The sample unit shall be one suit. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

DEFECTS

Any measurement deviating from any dimension specified in table II.

Sleeve lengths uneven by more than 1/2 inch.

Leg lengths uneven by more than 1/2 inch.

Specified number of desiccant units not enclosed.

4.4.5 End item testing. Samples of the suit shall be submitted to the Government for acceptance testing for the characteristic specified in table IV. The lot size shall be expressed in units of suits. The sample unit shall be one suit. Any test failure shall be cause for rejection of the lot. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

TABLE IV. End item testing

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported numerically to nearest
Carbon tetrachloride adsorption	3.10	1/	4 2/	0.1 mg/cm^2

1/ The test procedure shall be as specified in MIL-C-43858, except that the test samples shall be conditioned in an air circulating oven at 50°C for at least 3 hours and that no individual sample determination shall be less than 1.3 mg/cm^2.

2/ A sample of the seamless impregnated material shall be taken from an area adjacent to each of the following areas of the coat and trouser:

Coat a. At the underarm including sleeve, side and armhole seams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

	b. At the top of shoulder including shoulder, armhole and sleeve seams.
Trousers	a. At the crotch including crotch, seat, and inseams.
	b. At the side seam midway between top of waist and top of protective flap at side opening.

4.4.5.1 Carbon tetrachloride warning. Carbon tetrachloride is a suspect carcinogen per OSHA and the American Conference of Governmental Industrial Hygienists - Threshold Limit Value (ACGIH-TLV) Booklet (ACGIH, Cincinnati, OH, 1986, page 40). Personnel exposed to carbon tetrachloride should be informed of the potential hazards. Use should be contained in a ventilation hood. Ensure personnel use measures to prevent respiratory, skin, or other exposure by using personnel protective equipment. Consult the Material Safety Data Sheet (MSDS) for handling procedures and appropriate protective equipment.

4.4.6 Packaging examination.

4.4.6.1 Seam strength testing. Filled and sealed unit packs shall be tested as specified in ASTM F 88 for conformance to seam strength requirements in 5.1.1.1. Lot size shall be expressed in unit packs. The sample unit shall be one unit pack. The inspection level shall be S-2. Any failure shall result in rejection of the lot.

4.4.6.2 In-process unit pack examination. Every filled and sealed unit pack shall be examined for the defects listed below. The finding of any defect shall be cause for rejection of the unit pack. Suits from rejected unit packs may be re-unit packed as specified in 5.1.1.1 and submitted for reexamination.

Defect

Closure seal width not as specified.
Closure seal not continuous.
Tear, cut, or hole in unit pack.

4.4.6.3 Unit pack leakage testing. Every filled and sealed unit pack, prior to overpacking as specified in 5.1.1.2, shall be tested as specified in 4.5.1 for conformance to the leakage requirements in 5.1.1.1. Any unit pack which fails the test in 4.5.1 shall be opened and the suit shall be removed, thoroughly dried, re-unit packed as specified in 5.1.1.1, and retested as above.

4.4.6.4 Overpack volume testing. Filled and sealed overpacks shall be tested as specified in 4.5.2 for conformance to the volume requirements in 5.1.1.2. The lot size shall be expressed in overpacks. The sample unit shall be one overpack. The inspection level shall be S-2. Any failure shall result in rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

4.4.6.5 Overpack examination. Every filled and sealed overpack shall be examined for the defects listed below. The finding of any defect shall be cause for rejection of the overpack. Unit packs from rejected overpacks may be re-overpacked as specified in 5.1.1.2 and submitted for reexamination.

Defect
Closure seal width not as specified
Closure seal not continuous
Tear, cut, or hole in overpack

4.4.6.6 Shipping container examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container, fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or incomplete waterproofing. Bulged or distorted container. Inside dimensions of container not as specified. Partition not as specified.
Content	Number of unit packs per container less than required. Size shown on one or more unit packs not as specified on container. 1/

1/ For this defect, one unit pack from each container in the sample shall be examined.

4.4.6.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Leakage test. Each unit pack shall be tested for leaks in accordance with the hot water technique of Method 5009 of FED-STD-101. Bubbles which may appear on the surface of the unit pack but are not released at a slowly decreasing rate are not to be construed as an indication of failure. A steady stream or recurring succession of bubbles from any surface or seam shall be considered a failure. Upon completion of the test, the unit packs shall be thoroughly dried prior to overpacking.

4.5.2 Volume test. The total volume of the finished overpacked unit pack shall be determined by a water displacement method. The measuring container shall have a surface area of not more than 300 square inches. The container shall be filled with water to a level that allows total submersion of the overpack. A vertical gauge shall be mounted on the container so that the change in water level can be determined. Each finished overpacked unit pack shall be placed in the water and held under the water so that the top surface of the overbag is 1/8 to 2 inches below the surface of the water. When the water turbulence has subsided to less than 1/8 inch total fluctuation, the vertical gauge shall be read. Calculate the total displacement of water to determine volume (see 5.1.1.2).

4.5.3 Loose stitch test prior to daily production. All single stitching for attaching foam knit laminate cloth to itself shall be evaluated to assure that the seams are properly constructed so as to compensate for the foam compression. Visually, the overall seam structure shall be embedded into the face of the foam laminate. Maximum seam clearance shall be confirmed by the following apparatus and procedure:

a. Apparatus: Hand sewing needle, size 70/2 sewing thread conforming to A-A-50199, 2.0 ounces (57 $\pm$ 6 gm) weight including hook, fine grit sandpaper, thickness gage 0.1 psi having a 0.001 inch precision capability.

b. Procedure: Dull the tip of a typical hand sewing needle with fine grit sandpaper and feed three strands of 70/2 sewing thread through the needle's eye. Tie the total six strands into a single over hand knot about

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

6 inches from the needle's eye and tighten knot by hand until a knot size of 0.036 ± 0.002 inch is achieved at the thickest spot using the specified thickness gage for measurement. From the knit side of the foam/knit lamininate cloth, intersect the sewing needle between two stitches being careful not to catch either the knit laminate cloth or the stitch thread. With the foam side of the foam/knit laminate cloth facing down, place thumbs about 1/2 inch to either side of thread intersection and force the seam allowance foam sides to be flush with the base fabric foam side. Hook the intersecting thread on needle side to the 2 ounce weight and gently lift the foam/knit cloth upwards until the weight is free hanging (see figure 3). Allow weight to hang for 5 seconds. If weight remains hanging after the 5 seconds, then the seam shall be considered to be properly constructed. If the weight pulls the knot through the seam, the seam is too loose and considered a critical defect.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation.

5.1.1.1 Unit packing. One suit consisting of a coat and pair of trousers of the same size, cut and assembled from the same lot of impregnated cloth, and six desiccant units conforming to type II of MIL-D-3464 shall be unit packed in a notched, water vaporproof, greaseproof barrier bag conforming to type I, class E, style 1 of MIL-B-117. The suit shall be folded and then rolled together to form a cylindrical shape. During the process of rolling the suit, the six desiccant units shall be placed toward the center of the roll along with all grommets, buckles and slide fasteners. The total time of exposure of the desiccant to the atmosphere after removal from its sealed container until final closure of the barrier bag shall not exceed 2 hours. In order to maintain the cylindrical shape of the suit while packaging, the suit shall be secured with a minimum 2 inch wide, 50 pound basis weight kraft paper strips that are fastened together with pressure sensitive adhesive tape. The outer surfaces of the barrier bag shall be uniformly colored overall with a color in the range of 34079 to 34087, inclusive, of FED-STD-595. The barrier bag shall be flat style, having inside dimensions of 14-1/2 by 23 ± 1/4 inches for end opening bags and 20 by 17 ± 1/4 inches for side opening bags. Excess air in the bag shall be removed by physical compression or by use of a partial vacuum (not to exceed 15 inches of mercury). The closure shall be heat sealed. The heat seal shall be 3/16 to 1/2 inch in width, uniform and smooth (without wrinkles or foreign matter) and shall have a seam strength of not less than 7 pounds per inch of width when tested in accordance with 4.4.6.1. The filled and closed unit shall not leak when tested in accordance with 4.4.6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

5.1.1.2 Overpacking. Each suit, unit packed as specified in 5.1.1.1, shall be thoroughly dried after testing in 4.4.6.3 and overpacked in a plastic pouch conforming to size 1 of MIL-P-44406. The corners of the unit packed suit shall be folded onto itself, as necessary and in such a manner as not to puncture either the pouch material or the suit unit pack material and to facilitate the heat seal closing of the pouch. The large opening of the pouch shall than be closed using a continous heat seal. The heat seal shall be 3/16 to 1/2 inch in width, uniform and smooth (without wrinkles or foreign matter) and shall have a seam strength of not less than 7 pounds per inch of width when tested in accordance with ASTM F 88. A vacuum tube shall be inserted into the pouch through the vacuum port. Excess air trapped between the unit pack and the pouch shall be extracted using a vacuum. After removal of the excess air, the vacuum port of the pouch shall be pinched-off as a vacuum tube is being withdrawn and then immediately closed with a heat seal. The overpack shall have a total volume of not more than 800 cubic inches for XXX-small to medium suits and not more than 850 cubic inches for large to XX-large suits, when tested in accordance with 4.4.6.4.

5.2 Packing. Packing shall be level A or B as specified (see 6.2 and 6.5.2).

5.2.1 Level A packing. Four suits of one class, one size and one lot only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, class weather-resistant, grade V2s of PPP-B-636. Inside dimensions of each shipping container shall be in accordance with table V. If the box bottom has been stapled or stitched, the box shall be fitted with a bottom pad. Each shipping container shall be fitted with a full depth corrugated partition to create four equal sized cells. The partition and pad (if used), shall be made from fiberboard conforming to grade V3c of PPP-F-320. Each container shall be closed in accordance with method III, waterproofed and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four suits of one class, one size and one lot only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, class weather-resistant, grade V3c of PPP-B-636. Inside dimensions of each shipping container shall be in accordance with table V. If the box bottom has been stapled or stitched, the box shall be fitted with a bottom pad. Each shipping container shall be fitted with a full depth corrugated partition to create four equal sized cells. The partition and pad (if used), shall be made from fiberboard conforming to grade V3c of PPP-F-320. Each container shall be closed in accordance with method III and waterproofed in accordance with method V as specified in the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

TABLE V. Inside dimensions of shipping container

Suit size	Container size (inches)			
	Length	Width	Depth	Tolerance
XXX-small to medium	16	15-1/4	13-1/4	± 3/16
Large to XX-large	17	15-1/2	14-1/2	± 3/16

5.3 Palletization. When specified (see 6.2), suits packed as specified in 5.2.2 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 95 in accordance with appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129. Inspection/test data as specified in MIL-STD-129 for shelf life markings shall be based on the date of impregnation of the foam material and not the date that the suit is manufactured. (The last six digits of the surveillance number are the day, month, and year of impregnation.) Additionally, unit packs shall be marked as specified in 5.4.1, 5.4.2 and 5.4.3. All unit pack markings, except for the fit and size prediction legend, shall be marked on the same face of the vapor barrier bag. Size and fit legend shall be marked on the reverse face.

5.4.1 Surveillance marking.

5.4.1.1 Unit pack. Each unit pack shall be marked with the surveillance marking indicated on the inclosed suit (see 3.3.13.1). The markings shall be printed or stamped below the identification and contract data markings in 1/2 inch high black characters.

5.4.1.2 Shipping container. Each shipping container shall be marked with the surveillance marking indicated on the inclosed suit (see 3.3.13.1). The markings shall be stenciled, stamped or printed on the same side of the container as the identification and contract data markings and on one end of the container in 1-inch high black characters.

5.4.2 Fit and size prediction legend markings. Each unit pack shall be marked with a fit and size prediction legend as specified in 3.3.13.5. Markings shall be printed in black, bold face characters. The item identification characters shall be a minimum of 18 point (approximately 1/4 inch in height) and the instructive information characters shall be a minimum of 10 point (approximately 1/8 inch in height).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

5.4.3 Special marking.

5.4.3.1 Unit pack. Each unit pack shall be marked in black with a capital letter "C", for class 1, Woodland Camouflage and a capital letter "D-3" for class 3, Desert Camouflage (3 color). This letter shall be 4 ± 1 inches in height and centered on the face of the vapor barrier bag.

5.4.3.2 Shipping container. Each shipping container shall be marked in black on three sides with the characters "4C", for class 1, Woodland Camouflage and "4D-3" for class 3 Desert Camouflage (3 color). These characters shall be 4 ± 1 inches in height and centered on the three sides of the shipping container. The side containing the identification and contract data marking need not be marked.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The suit is intended for use by combat personnel to provide protection against battlefield concentrations of chemical agents in liquid, vapor, or aerosol form.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Class and size (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.1, 4.3, and 6.3).
f. Level of packing (see 5.2).
g. Type and class of unit load required (see 5.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample and shade sample. For access to sample of the end item and samples for shade, address the contracting activity issuing the invitation for bids or request for proposal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

6.5 Warning.

6.5.1 Protection of processed goods. Processed goods should be protected from exposure to chemical vapors such as solvents to prevent contamination of the activated carbon.

6.5.2 Personnel protection. During the handling of the charcoal containing material used in fabricating the Suit, Chemical Protective, charcoal from the material may collect in various places; for example, on the skin of the personnel, particularly on the hands, and on sewing machines and cutting tables. Therefore, it is essential that the industrial hygiene procedures required by the Occupational Safety and Health Standards be closely adhered to during the manufacturing of the suit. Provide general and local exhaust ventilation sufficient to maintain airborne dust concentration below that listed in OSHA 1910.1000. Otherwise, respiratory protection may be required due to charcoal dust generation. Personnel should wear gloves and protective clothing when handling the material. Chemical safety goggles should be worn whenever eye contact may occur. Personnel should not wear contact lens. Also, washing facilities and eyewash stations should be available in the work areas.

6.6 Subject (key word) listing.

Chemical warfare agent
Combat
Disposable
Expendable clothing
Protection

6.7 Supersession data. The class 2 desert camouflage pattern (6 color) suit has been deleted as it is no longer required.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD, EA
Air Force - 11, 82
DLA-CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

(Project 8415-0721)

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:01Z
Check the source to verify that this is the current version before use.

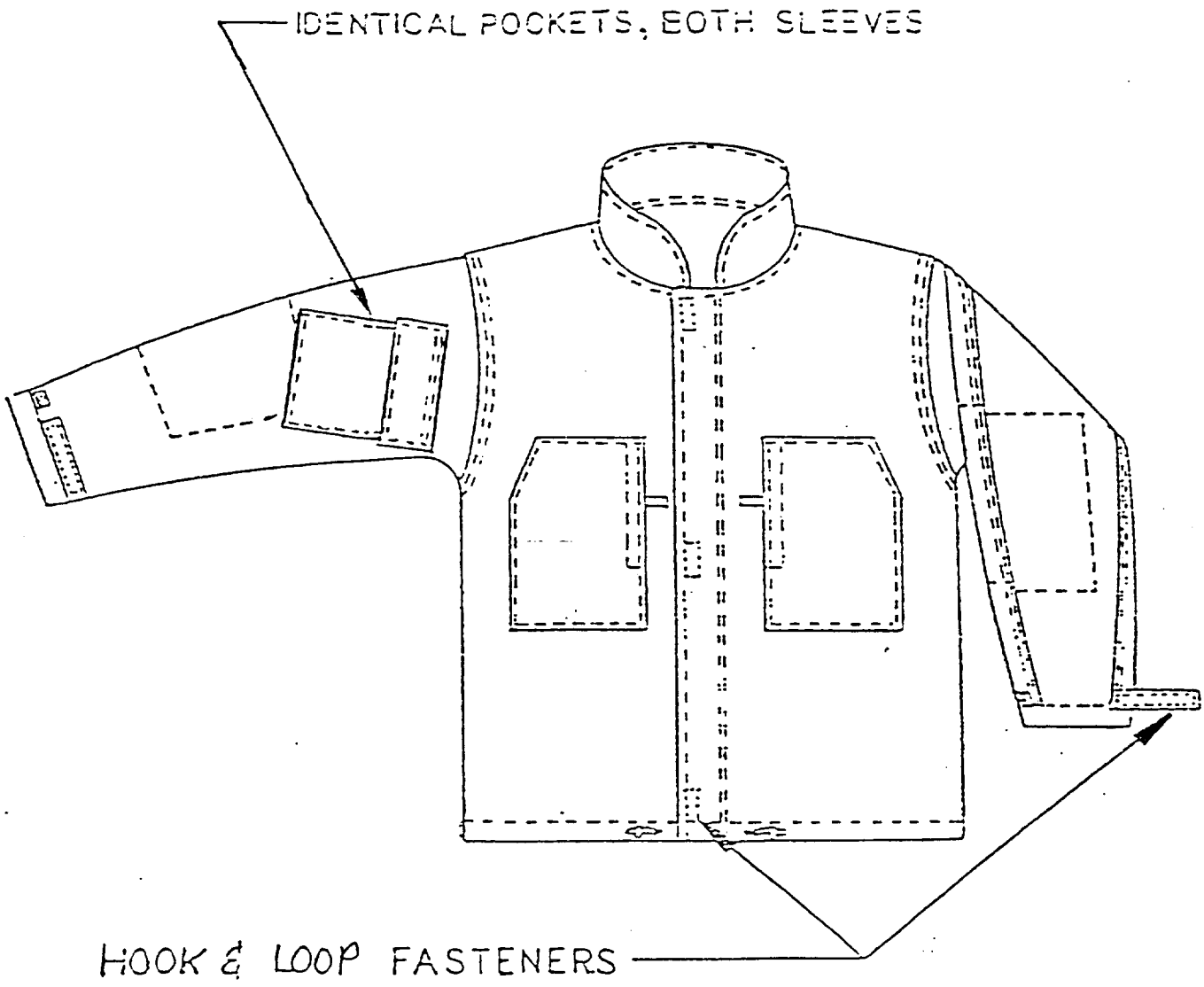
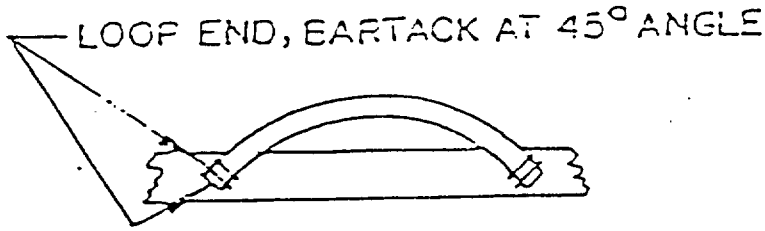


FIG-1- SUIT, CHEMICAL PROTECTIVE (COAT)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.



SUSPENDER LOOP ATTACHMENT

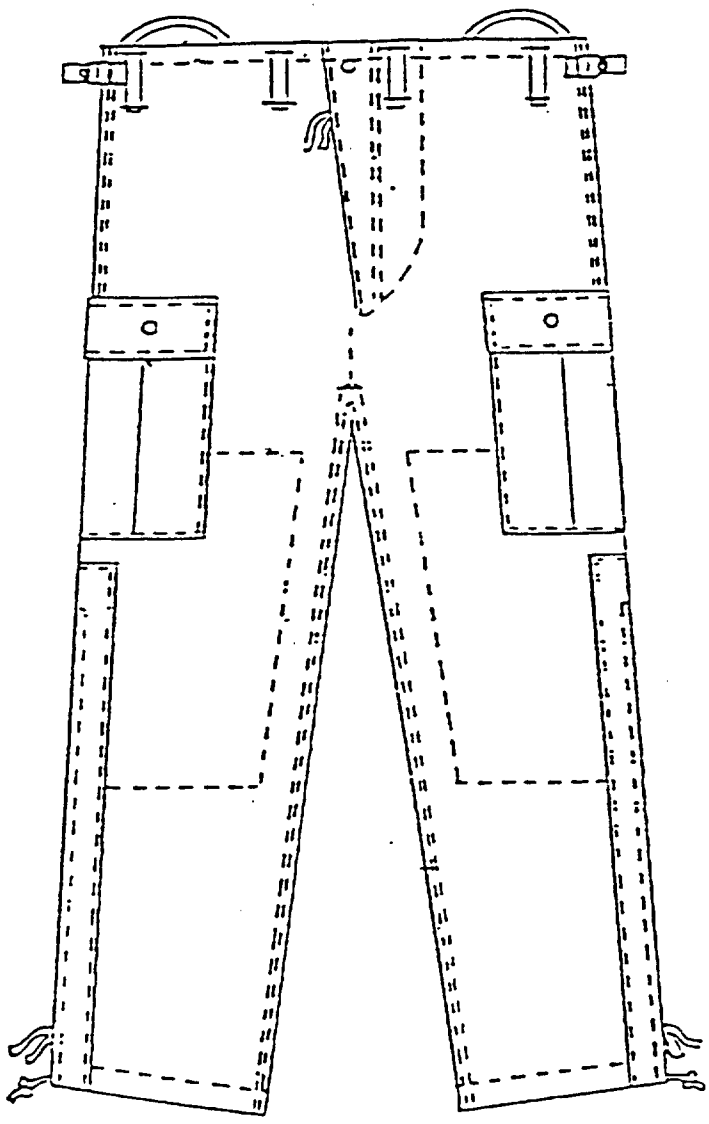


FIG-2- SUIT, CHEMICAL PROTECTIVE (TROUSERS)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

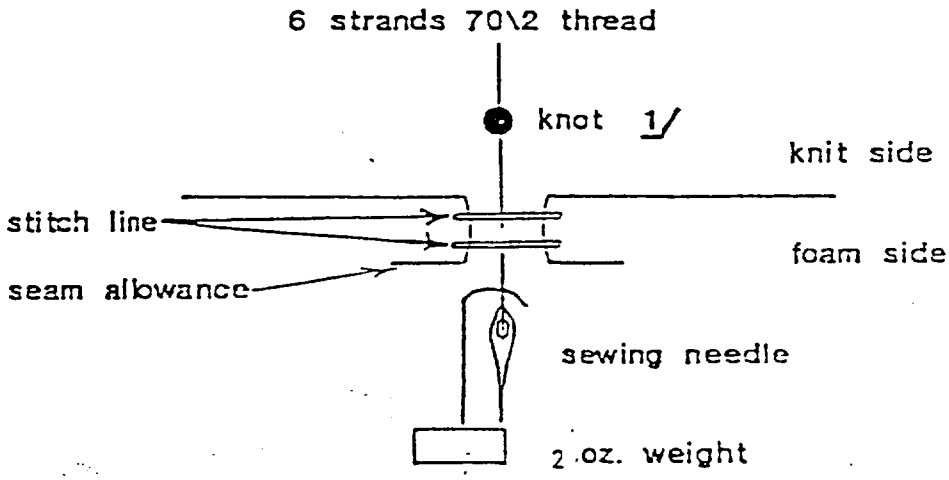


Fig. 3 - Loose stitch test

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-43926J	1991 August 30

3. DOCUMENT TITLE

SUIT, CHEMICAL PROTECTIVE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 20240-21123:01Z
Check the source to verify that this is the current version before use.